Volume II

Origins of Numbers

The Continuum

Self-Study Notes

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From Cantor to Ito

Front Matter

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Book. The Continuum

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Chapter 1

Embedding the Number Systems

embedding-number-systems

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1.1 Embeddings as Structure Preserving Maps

From Construction to Identification

The preceding chapters built $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ one at a time. In that construction-first development the symbol 2 denotes a different object in each system: a Peano element, a whole number, an integer difference object, a rational fraction object, and a real number are literally distinct sets. Nothing in the constructions has made them identical.

An *embedding* is the device that closes this gap without pretending the objects were the same to begin with. It is an injective map that preserves the relevant structure present in the source system — the distinguished constants, the operations, and the order being considered. Once such a map is fixed, the source system is identified with its image inside the target, and only then is the familiar chain $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ written down. The inclusion symbol records an identification along an embedding, not a literal subset relation.

Definition (Structure-Preserving Map)

Definition 1.1.1 (Structure-Preserving Map)**.** Let A and B be ordered arithmetic systems with a specified common source structure: selected constants, selected operations, and a strict order. A map $\varphi : A \to B$ is *structure-preserving* exactly when it preserves each selected constant and, for every selected binary operation $\star$ and all $a, a' \in A$,

$$\varphi(a \star_A a') = \varphi(a) \star_B \varphi(a'),$$

and $a <_A a' \implies \varphi(a) <_B \varphi(a')$.

Remark (Standard quantified statement).

$$\begin{aligned}&\varphi : A \to B \text{ is structure-preserving}\\ \iff &\Big(\text{every selected constant is preserved}\\ &\wedge\ \text{every selected operation is preserved}\\ &\wedge \forall a, a' \in A\ (a <_A a' \Rightarrow \varphi(a) <_B \varphi(a'))\Big).\end{aligned}$$

Remark (Interpretation). A structure-preserving map carries the chosen arithmetic and order from A into B without distortion. The choice of structure matters: the positive part of $\mathbb{W}$ carries the Peano successor from $\mathbb{N}$, while the later ring and field embeddings focus on 0, 1, addition, multiplication, and order. A structure-preserving map need not be injective; collapsing distinct elements is still allowed at this stage. Injectivity is added separately to obtain an embedding.

Remark (Dependencies).

- Strict Total Order
- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

Definition (Order Embedding)

Definition (Order Embedding)**.** Let A and B be ordered arithmetic systems and $\varphi : A \to B$ a map. Then φ is an *order embedding* exactly when, for all $a, a' \in A$,

$$a <_A a' \iff \varphi(a) <_B \varphi(a').$$

Definition (Embedding of Number Systems)

Definition 1.1.2 (Embedding of Number Systems)**.** Let A and B be ordered arithmetic systems. A map $\varphi : A \to B$ is an *embedding* exactly when φ is injective and structure-preserving and is an order embedding.

Remark (Standard quantified statement).

$$\begin{aligned}&\varphi \text{ is an embedding}\\ \iff &\Big(\varphi \text{ is injective} \wedge \varphi \text{ is structure-preserving}\\ &\wedge \forall a, a' \in A\ (a <_A a' \Leftrightarrow \varphi(a) <_B \varphi(a'))\Big).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\varphi \text{ is not an embedding}\\ \iff &\Big(\exists a, a' \in A\ (a \neq a' \wedge \varphi(a) = \varphi(a'))\\ &\vee\ \varphi \text{ fails to preserve a constant, sum, or product}\\ &\vee\ \exists a, a' \in A\ \big((a <_A a' \wedge \neg(\varphi(a) <_B \varphi(a'))) \vee (\varphi(a) <_B \varphi(a') \wedge \neg(a <_A a'))\big)\Big).\end{aligned}$$

Remark (Failure modes). An embedding can fail in three structurally distinct ways: it may collapse two distinct elements to the same image (injectivity fails), fail to transport one of the selected constants or operations (algebraic preservation fails), or fail to preserve or reflect a strict-order comparison (order embedding fails).

Remark (Interpretation). An embedding is the precise meaning of "A sits inside B." It is injective, so distinct elements stay distinct; it preserves the arithmetic and reflects the order, so the copy of A inside B behaves exactly as A did. This is what licenses treating $2 \in \mathbb{N}$ and $2 \in \mathbb{R}$ as "the same number."

Remark (Interpretation). Most embeddings after $\mathbb{W}$ transport the *ordered ring/field* structure: constants, the two operations, and order. The first bridge $\mathbb{N} \hookrightarrow \mathbb{W}$ is slightly different because $\mathbb{N}$ is one-based and carries Peano successor. That local theorem is proved in the whole-number chapter. In every case, induction is not transported to the larger target: the phrase "the naturals sit inside the integers" must not be read as carrying induction across all of $\mathbb{Z}$.

Remark (Dependencies).

- Structure-Preserving Map
- Order Embedding

Definition (Image of an Embedding)

Definition 1.1.3 (Image of an Embedding)**.** Let $\varphi : A \to B$ be an embedding. The *image* of φ is the set $\varphi(A) = \{\, \varphi(a) : a \in A \,\} \subseteq B$, equipped with the selected constants, operations, and order of B restricted to $\varphi(A)$.

Remark (Standard quantified statement).

$$\varphi(A) = \{\, b \in B : \exists a \in A \; (b = \varphi(a)) \,\}.$$

Remark (Interpretation). The image is the literal subset of B that plays the role of A. The identification theorem below states that φ, viewed as a map onto its image, is an isomorphism, so A and $\varphi(A)$ are the same ordered arithmetic system under two names.

Remark (Dependencies).

- Embedding of Number Systems

Theorem (Embeddings Identify a System with Its Image)

Theorem 1.1.4 (Embeddings Identify a System with Its Image)**.** *Let $\varphi : A \to B$ be an embedding of ordered arithmetic systems relative to a selected source structure. Then the corestriction $\varphi : A \to \varphi(A)$ is an isomorphism onto its image: it is a bijection that preserves and reflects the selected constants, operations, and order. Consequently A may be identified with the corresponding substructure $\varphi(A) \subseteq B$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\varphi : A \to B \text{ an embedding} \\ &\implies \varphi : A \to \varphi(A) \text{ is a bijective isomorphism for the selected structure.} \end{aligned}$$

Remark (Interpretation). This is the theorem that makes the inclusion notation legitimate. After identification, $A \subseteq B$ abbreviates "A embeds via φ as the substructure $\varphi(A)$." Every later use of $\mathbb{N} \subseteq \mathbb{R}$ is shorthand for this identification.

Remark (Dependencies).

- Embedding of Number Systems
- Image of an Embedding

1.2 Embedding N Into W

Embedding $\mathbb{N}$ into $\mathbb{W}$

This section records, from the global embedding viewpoint, the local bridge proved in the whole-number chapter: $\mathbb{N}$ sits inside $\mathbb{W}$ as the positive part. The map is the bridge that lets elements of $\mathbb{N}$ be regarded as nonzero elements of $\mathbb{W}$ after identification.

Definition (Embedding $\mathbb{N}$ into $\mathbb{W}$)

Definition 1.2.1 (Embedding $\mathbb{N}$ into $\mathbb{W}$)**.** The *canonical embedding* $\iota : \mathbb{N} \to \mathbb{W}$ is defined by

$$\iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Remark (Standard quantified statement).

$$\iota : \mathbb{N} \to \mathbb{W}, \qquad \iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Remark (Interpretation). The map sends each element of $\mathbb{N}$ to its natural representative in $\mathbb{W}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{N}$ may be identified with its image in $\mathbb{W}$.

Remark (Dependencies).

- Construction of $\mathbb{W}$ from $\mathbb{N}$ by adjoining 0
- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

Theorem ($\mathbb{N}$ Embeds in $\mathbb{W}$)

Theorem 1.2.2 ($\mathbb{N}$ Embeds in $\mathbb{W}$)**.** *The canonical map $\iota : \mathbb{N} \to \mathbb{W}$ identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{N} \to \mathbb{W} \setminus \{0\} \text{ is bijective and preserves successor, arithmetic, and order.}$$

Remark (Interpretation). Once this theorem is in place, $\mathbb{N}$ may be treated as the positive substructure $\iota(\mathbb{N}) = \mathbb{W} \setminus \{0\}$ inside $\mathbb{W}$. The new element of $\mathbb{W}$ is exactly the adjoined zero.

Remark (Dependencies).

- Construction of $\mathbb{W}$ from $\mathbb{N}$ by adjoining 0
- Embedding $\mathbb{N}$ into $\mathbb{W}$
- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part

1.3 Embedding W Into Z

Embedding $\mathbb{W}$ into $\mathbb{Z}$

This section fixes the canonical embedding of $\mathbb{W}$ into $\mathbb{Z}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{W}$ be regarded as elements of $\mathbb{Z}$ after identification.

Definition (Embedding $\mathbb{W}$ into $\mathbb{Z}$)

Definition 1.3.1 (Embedding $\mathbb{W}$ into $\mathbb{Z}$)**.** The *canonical embedding* $\iota : \mathbb{W} \to \mathbb{Z}$ is defined by

$$\iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0\text{)}.$$

Remark (Standard quantified statement).

$$\iota : \mathbb{W} \to \mathbb{Z}, \qquad \iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0\text{)}.$$

Remark (Interpretation). The map sends each element of $\mathbb{W}$ to its natural representative in $\mathbb{Z}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{W}$ may be identified with its image in $\mathbb{Z}$.

Remark (Dependencies).

- Formal-Difference Equivalence Is an Equivalence Relation
- Embedding of Number Systems

Theorem ($\mathbb{W}$ Embeds in $\mathbb{Z}$)

Theorem 1.3.2 ($\mathbb{W}$ Embeds in $\mathbb{Z}$)**.** *The canonical map $\iota : \mathbb{W} \to \mathbb{Z}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{W} \to \mathbb{Z} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation). Once this theorem is proved, the identification theorem of §1.1 applies, and $\mathbb{W}$ may be treated as the substructure $\iota(\mathbb{W}) \subseteq \mathbb{Z}$.

Remark (Dependencies).

- Formal-Difference Equivalence Is an Equivalence Relation
- Embedding $\mathbb{W}$ into $\mathbb{Z}$
- Embeddings Identify a System with Its Image

1.4 Embedding Z Into Q

Embedding $\mathbb{Z}$ into $\mathbb{Q}$

This section fixes the canonical embedding of $\mathbb{Z}$ into $\mathbb{Q}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{Z}$ be regarded as elements of $\mathbb{Q}$ after identification.

Definition (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)

Definition 1.4.1 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)**.** The *canonical embedding* $\iota : \mathbb{Z} \to \mathbb{Q}$ is defined by

$$\iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1).$$

Remark (Standard quantified statement).

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1).$$

Remark (Interpretation)*.* The map sends each element of $\mathbb{Z}$ to its natural representative in $\mathbb{Q}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{Z}$ may be identified with its image in $\mathbb{Q}$.

Remark (Dependencies).

- Construction of $\mathbb{Q}$ as fraction classes of integers
- Embedding of Number Systems

Theorem ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)

Theorem 1.4.2 ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)**.** *The canonical map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{Z} \to \mathbb{Q} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation)*.* Once this theorem is proved, the identification theorem of §1.1 applies, and $\mathbb{Z}$ may be treated as the substructure $\iota(\mathbb{Z}) \subseteq \mathbb{Q}$.

Remark (Dependencies).

- Construction of $\mathbb{Q}$ as fraction classes of integers
- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Embeddings Identify a System with Its Image

1.5 Embedding Q Into R

Embedding $\mathbb{Q}$ into $\mathbb{R}$

This section fixes the canonical embedding of $\mathbb{Q}$ into $\mathbb{R}$ and records that it is injective, structure-preserving, and order-reflecting. The map is the bridge that lets elements of $\mathbb{Q}$ be regarded as elements of $\mathbb{R}$ after identification.

Definition (Embedding $\mathbb{Q}$ into $\mathbb{R}$)

Definition 1.5.1 (Embedding $\mathbb{Q}$ into $\mathbb{R}$)**.** The *canonical embedding* $\iota : \mathbb{Q} \to \mathbb{R}$ is defined by

$$\iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Remark (Standard quantified statement).

$$\iota : \mathbb{Q} \to \mathbb{R}, \qquad \iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Remark (Interpretation). The map sends each element of $\mathbb{Q}$ to its natural representative in $\mathbb{R}$. The next result certifies that this representative choice preserves all arithmetic and order, so $\mathbb{Q}$ may be identified with its image in $\mathbb{R}$.

Remark (Dependencies).

- Construction of $\mathbb{R}$ by Dedekind cuts
- Embedding of Number Systems

Theorem ($\mathbb{Q}$ Embeds in $\mathbb{R}$)

Theorem 1.5.2 ($\mathbb{Q}$ Embeds in $\mathbb{R}$)**.** *The canonical map $\iota : \mathbb{Q} \to \mathbb{R}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order. Go to proof.*

Remark (Standard quantified statement).

$$\iota : \mathbb{Q} \to \mathbb{R} \text{ is injective, structure-preserving, and order-reflecting.}$$

Remark (Interpretation). Once this theorem is proved, the identification theorem of §1.1 applies, and $\mathbb{Q}$ may be treated as the substructure $\iota(\mathbb{Q}) \subseteq \mathbb{R}$.

Remark (Dependencies).

- Construction of $\mathbb{R}$ by Dedekind cuts
- Embedding $\mathbb{Q}$ into $\mathbb{R}$
- Embeddings Identify a System with Its Image

1.6 Compatibility of Arithmetic and Order

One Arithmetic Through the Chain

Each embedding preserves arithmetic and order individually. What remains is to confirm that the embeddings are mutually compatible: composing $\mathbb{N} \hookrightarrow \mathbb{W} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}$ agrees with the direct embedding of $\mathbb{N}$ into $\mathbb{R}$, so that "2" has one unambiguous meaning after all identifications.

Theorem (Compatibility of the Embedding Chain)

Theorem 1.6.1 (Compatibility of the Embedding Chain)**.** *Let $\iota_1 : \mathbb{N} \to \mathbb{W}$, $\iota_2 : \mathbb{W} \to \mathbb{Z}$, $\iota_3 : \mathbb{Z} \to \mathbb{Q}$, and $\iota_4 : \mathbb{Q} \to \mathbb{R}$ be the canonical embeddings. For every $n \in \mathbb{N}$ the composite $(\iota_4 \circ \iota_3 \circ \iota_2 \circ \iota_1)(n)$ is the real number identified with n, and the composite preserves the arithmetic and order carried by the source system. Hence arithmetic computed in any system agrees with arithmetic on its image in $\mathbb{R}$. Go to proof.*

Remark (Standard quantified statement).

$$\forall n, m \in \mathbb{N} \ \big(\Phi(n{+}m) = \Phi(n){+}\Phi(m) \wedge \Phi(n{\cdot}m) = \Phi(n){\cdot}\Phi(m) \wedge (n < m \Leftrightarrow \Phi(n) < \Phi(m))\big), \quad \Phi = \iota_4{\circ}\iota_3{\circ}$$

Remark (Interpretation). Compatibility is what makes the single symbol 2 well-defined across the whole chain: the answer to $2 + 2$ is the same whether computed in $\mathbb{N}$ or after embedding into $\mathbb{R}$. Without this coherence the inclusion notation would silently carry two different meanings for the same numeral.

Remark (Dependencies).

- $\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part
- $\mathbb{W}$ Embeds in $\mathbb{Z}$
- $\mathbb{Z}$ Embeds in $\mathbb{Q}$
- $\mathbb{Q}$ Embeds in $\mathbb{R}$

1.7 One Chain Many Structures

One Chain, Many Structures

The embeddings assemble into a single chain $\mathbb{N} \hookrightarrow \mathbb{W} \hookrightarrow \mathbb{Z} \hookrightarrow \mathbb{Q} \hookrightarrow \mathbb{R}$. Each arrow is injective, preserves the relevant arithmetic, and reflects order; each step also *adds* structure that the previous system lacked.

From $\mathbb{N}$ to $\mathbb{W}$ an additive identity 0 appears. From $\mathbb{W}$ to $\mathbb{Z}$ additive inverses appear, completing a commutative ring. From $\mathbb{Z}$ to $\mathbb{Q}$ multiplicative inverses of nonzero elements appear, completing an ordered field. From $\mathbb{Q}$ to $\mathbb{R}$ order-completeness appears: the gaps in the rational line are filled. The numerals carry through unchanged by compatibility, but the ambient structure grows at every step. This chain is the bridge from number construction to the analysis of the real line.

Remark (Exposition). Read as a ladder of structures: counting ($\mathbb{N}$) $\to$ counting with zero ($\mathbb{W}$) $\to$ signed counting / ring ($\mathbb{Z}$) $\to$ ordered field ($\mathbb{Q}$) $\to$ complete ordered field ($\mathbb{R}$). Only the last rung is order-complete, and that completeness is the property the structure-of-the-real-line chapter turns into metric topology.

Chapter 2

Rational Numbers ($\mathbb{Q}$)

rationals

Integers ($\mathbb{Z}$) $\rightarrow$ **Rationals ($\mathbb{Q}$)** $\rightarrow$ Real Numbers ($\mathbb{R}$)

2.1 Construction of $\mathbb{Q}$

Prefractions

Definition

Definition 2.1.1 (Nonzero Integers)**.** The set of nonzero integers is

$$\mathbb{Z}^* := \mathbb{Z} \setminus \{0_\mathbb{Z}\}.$$

Remark (Dependencies).

- Integers
- Zero in $\mathbb{Z}$

Definition

Definition 2.1.2 (Prefraction)**.** A *prefraction* is an ordered pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as the formal quotient a/b.

Remark (Dependencies).

- Ordered Pair
- Integers
- Nonzero Integers

Definition

Definition 2.1.3 (Prefraction Set)**.** The prefraction set is

$$\operatorname{Pre}\mathbb{Q} := \mathbb{Z} \times \mathbb{Z}^*.$$

Remark (Dependencies).

- Cartesian Product
- Integers
- Nonzero Integers

Fraction Equivalence

Definition

Definition 2.1.4 (Fraction Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$, define

$$(a, b) \sim_{\mathbb{Q}} (c, d) \quad :\Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Remark (Dependencies)*.*

- Relation
- Prefraction Set
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 2.1.5 (Fraction Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Q}$*,*

$$(a, b) \sim_{\mathbb{Q}} (a, b).$$

Remark (Dependencies)*.*

- Fraction Equivalence
- Multiplication on $\mathbb{Z}$ Is Commutative

Lemma

Lemma 2.1.6 (Fraction Equivalence Is Symmetric)**.** *For all prefractions* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \implies (c, d) \sim_{\mathbb{Q}} (a, b).$$

Remark (Dependencies)*.*

- Fraction Equivalence
- Multiplication on $\mathbb{Z}$ Is Commutative

Lemma

Lemma 2.1.7 (Fraction Equivalence Is Transitive)**.** *For all prefractions* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \land (c, d) \sim_{\mathbb{Q}} (e, f) \implies (a, b) \sim_{\mathbb{Q}} (e, f).$$

Remark (Dependencies)*.*

- Multiplicative Cancellation on $\mathbb{Z}$

Theorem

Theorem 2.1.8 (Fraction Equivalence Is an Equivalence Relation). *The relation* $\sim_{\mathbb{Q}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Q}$.

Remark (Dependencies).

- Equivalence Relation
- Fraction Equivalence
- Fraction Equivalence Is Reflexive
- Fraction Equivalence Is Symmetric
- Fraction Equivalence Is Transitive

Definition of $\mathbb{Q}$

Definition

Definition 2.1.9 (Rational Numbers). The rational numbers are the quotient

$$\mathbb{Q} := \operatorname{Pre}\mathbb{Q}/\sim_{\mathbb{Q}}.$$

Remark (Dependencies).

- Quotient Set
- Prefraction Set
- Fraction Equivalence

Definition

Definition 2.1.10 (Rational Class Notation). The notation $[a, b]$ denotes the $\sim_{\mathbb{Q}}$-equivalence class of the prefraction (a, b), where $b \neq 0_{\mathbb{Z}}$.

Remark (Dependencies).

- Rational Numbers
- Fraction Equivalence

Proposition

Proposition 2.1.11 (Equality Criterion for Rational Classes). *For prefractions* $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$,

$$[a, b] = [c, d] \quad \Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Remark (Dependencies).

- Rational Class Notation
- Fraction Equivalence
- Quotient Set
- Equivalence Class

Zero and One in $\mathbb{Q}$

Definition

Definition 2.1.12 (Zero in $\mathbb{Q}$)**.** Define

$$0_{\mathbb{Q}} := [0_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Class Notation
- Zero in $\mathbb{Z}$
- One in $\mathbb{Z}$

Definition

Definition 2.1.13 (One in $\mathbb{Q}$)**.** Define

$$1_{\mathbb{Q}} := [1_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Class Notation
- One in $\mathbb{Z}$

Lemma

Lemma 2.1.14 (Vanishing Criterion in $\mathbb{Q}$)**.** *For every prefraction* $(a, b) \in \operatorname{Pre}\mathbb{Q}$,

$$[a, b] = 0_{\mathbb{Q}} \iff a = 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Rational Class Notation
- Fraction Equivalence
- Zero in $\mathbb{Q}$
- Zero in $\mathbb{Z}$

Theorem

Theorem 2.1.15 (Zero Is a Rational)**.** *The class* $0_{\mathbb{Q}}$ *is an element of* $\mathbb{Q}$.

Remark (Dependencies).

- Rational Numbers
- Zero in $\mathbb{Q}$
- Rational Class Notation

Theorem

Theorem 2.1.16 (One Is a Rational)**.** *The class* $1_{\mathbb{Q}}$ *is an element of* $\mathbb{Q}$.

Remark (Dependencies).

- Rational Numbers
- One in $\mathbb{Q}$
- Rational Class Notation

Theorem

Theorem 2.1.17 (Zero Is Not One in $\mathbb{Q}$)**.** *In* $\mathbb{Q}$,

$$0_{\mathbb{Q}} \neq 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Zero Is Not One in $\mathbb{Z}$

2.2 Operations on $\mathbb{Q}$-Classes

Addition on $\mathbb{Q}$-Classes

Definition

Definition 2.2.1 (Addition on $\mathbb{Q}$-Classes)**.** Define addition on representatives by

$$[a, b] + [c, d] := [a \cdot d + b \cdot c,\ b \cdot d].$$

Remark (Dependencies).

- Rational Class Notation
- Addition on $\mathbb{Z}$
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 2.2.2 (Addition Lands in Pre $\mathbb{Q}$)**.** *If* $b \neq 0_{\mathbb{Z}}$ *and* $d \neq 0_{\mathbb{Z}}$, *then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Has No Zero Divisors

Lemma

Lemma 2.2.3 (Addition Respects Fraction Equivalence on the Left)**.** *If* $[a, b] = [a', b']$, *then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 2.2.4 (Addition Respects Fraction Equivalence on the Right)**.** *If* $[c, d] = [c', d']$, *then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 2.2.5 (Addition Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$, *then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition Respects Fraction Equivalence on the Left
- Addition Respects Fraction Equivalence on the Right

Theorem

Theorem 2.2.6 (Addition on $\mathbb{Q}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition Respects Fraction Equivalence

Theorem

Theorem 2.2.7 (Addition on $\mathbb{Q}$ Is Associative)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$(x + y) + z = x + (y + z).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Z}$ Is Associative
- Multiplication on $\mathbb{Z}$ Is Associative
- Multiplication Distributes over Addition on $\mathbb{Z}$

Theorem

Theorem 2.2.8 (Addition on $\mathbb{Q}$ Is Commutative)**.** *For all $x, y \in \mathbb{Q}$,*

$$x + y = y + x.$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 2.2.9 (Zero Is a Left Additive Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x.$$

Remark (Dependencies).

- Zero in $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Zero Is an Additive Identity on $\mathbb{Z}$
- One Is a Multiplicative Identity on $\mathbb{Z}$

Corollary

Corollary 2.2.10 (Zero Is a Right Additive Identity on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$,

$$x + 0_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.2.11 (Zero Is an Additive Identity on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$,

$$0_{\mathbb{Q}} + x = x \qquad \text{and} \qquad x + 0_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- Zero Is a Left Additive Identity on $\mathbb{Q}$
- Zero Is a Right Additive Identity on $\mathbb{Q}$

Negation on $\mathbb{Q}$-Classes

Definition

Definition 2.2.12 (Negation on $\mathbb{Q}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [-a, b].$$

Remark (Dependencies).

- Rational Class Notation
- Negation on $\mathbb{Z}$

Lemma

Lemma 2.2.13 (Negation Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$, *then*

$$-[a, b] = -[a', b'].$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Fraction Equivalence
- Sign Rule on $\mathbb{Z}$

Theorem

Theorem 2.2.14 (Negation on $\mathbb{Q}$ Is Well-Defined). *Negation determines a well-defined unary operation $\mathbb{Q} \to \mathbb{Q}$.*

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Negation Respects Fraction Equivalence

Theorem

Theorem 2.2.15 (Negation Gives a Left Additive Inverse on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$(-x) + x = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Negation on $\mathbb{Q}$ Is Well-Defined
- Addition on $\mathbb{Q}$ Is Well-Defined
- Negation Gives a Left Additive Inverse on $\mathbb{Z}$

Corollary

Corollary 2.2.16 (Negation Gives a Right Additive Inverse on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x + (-x) = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation Gives a Left Additive Inverse on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.2.17 (Every Rational Has an Additive Inverse). *For every $x \in \mathbb{Q}$ there exists $y \in \mathbb{Q}$ such that*

$$x + y = 0_{\mathbb{Q}} \qquad \text{and} \qquad y + x = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Negation on $\mathbb{Q}$ Is Well-Defined
- Negation Gives a Left Additive Inverse on $\mathbb{Q}$
- Negation Gives a Right Additive Inverse on $\mathbb{Q}$

Subtraction on $\mathbb{Q}$

Definition

Definition 2.2.18 (Subtraction on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x - y := x + (-y).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Negation on $\mathbb{Q}$

Theorem

Theorem 2.2.19 (Subtraction on $\mathbb{Q}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.*

Remark (Dependencies).

- Subtraction on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Negation on $\mathbb{Q}$ Is Well-Defined

Multiplication on $\mathbb{Q}$-Classes

Definition

Definition 2.2.20 (Multiplication on $\mathbb{Q}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [a \cdot c,\ b \cdot d].$$

Remark (Dependencies).

- Rational Class Notation
- Multiplication on $\mathbb{Z}$

Lemma

Lemma 2.2.21 (Multiplication Lands in Pre $\mathbb{Q}$)**.** *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Has No Zero Divisors

Lemma

Lemma 2.2.22 (Multiplication Respects Fraction Equivalence on the Left). *If* $[a, b] = [a', b']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 2.2.23 (Multiplication Respects Fraction Equivalence on the Right). *If* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Fraction Equivalence
- Equality Criterion for Rational Classes
- Multiplicative Cancellation on $\mathbb{Z}$

Lemma

Lemma 2.2.24 (Multiplication Respects Fraction Equivalence). *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication Respects Fraction Equivalence on the Left
- Multiplication Respects Fraction Equivalence on the Right

Theorem

Theorem 2.2.25 (Multiplication on $\mathbb{Q}$ Is Well-Defined). *Multiplication determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$

- Multiplication Respects Fraction Equivalence

Theorem

Theorem 2.2.26 (Multiplication on $\mathbb{Q}$ Is Associative). *For all $x, y, z \in \mathbb{Q}$,*

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Z}$ Is Associative

Theorem

Theorem 2.2.27 (Multiplication on $\mathbb{Q}$ Is Commutative). *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = y \cdot x.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Z}$ Is Commutative

Theorem

Theorem 2.2.28 (One Is a Left Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- One Is a Multiplicative Identity on $\mathbb{Z}$

Corollary

Corollary 2.2.29 (One Is a Right Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x \cdot 1_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Q}$

- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.2.30 (One Is a Multiplicative Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x \qquad \textit{and} \qquad x \cdot 1_{\mathbb{Q}} = x.$$

Remark (Dependencies).

- One Is a Left Multiplicative Identity on $\mathbb{Q}$
- One Is a Right Multiplicative Identity on $\mathbb{Q}$

Reciprocal on Nonzero $\mathbb{Q}$-Classes

Lemma

Lemma 2.2.31 (Nonzero Predicate Is Well-Defined on $\mathbb{Q}$)**.** *For every prefraction $(a, b) \in \operatorname{Pre}\mathbb{Q}$,*

$$[a, b] \neq 0_{\mathbb{Q}} \iff a \neq 0_{\mathbb{Z}}.$$

Thus nonzeroness is independent of the representative.

Remark (Dependencies).

- Zero in $\mathbb{Q}$
- Vanishing Criterion in $\mathbb{Q}$

Definition

Definition 2.2.32 (Reciprocal on Nonzero $\mathbb{Q}$-Classes)**.** For $[a, b] \neq 0_{\mathbb{Q}}$, define

$$[a, b]^{-1} := [b, a].$$

Remark (Dependencies).

- Rational Class Notation
- Vanishing Criterion in $\mathbb{Q}$

Lemma

Lemma 2.2.33 (Reciprocal Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If $[a, b] \neq 0_{\mathbb{Q}}$, then $a \neq 0_{\mathbb{Z}}$, so (b, a) is a prefraction.*

Remark (Dependencies).

- Prefraction
- Prefraction Set
- Vanishing Criterion in $\mathbb{Q}$

- Zero in $\mathbb{Z}$

Lemma

Lemma 2.2.34 (Reciprocal Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[a, b] \neq 0_{\mathbb{Q}}$*, then*

$$[a, b]^{-1} = [a', b']^{-1}.$$

Remark (Dependencies)*.*

- Reciprocal on $\mathbb{Q}$
- Fraction Equivalence
- Reciprocal Lands in Pre $\mathbb{Q}$

Theorem

Theorem 2.2.35 (Reciprocal on $\mathbb{Q}$ Is Well-Defined)**.** *Reciprocal determines a well-defined map*

$$\mathbb{Q} \setminus \{0_{\mathbb{Q}}\} \to \mathbb{Q} \setminus \{0_{\mathbb{Q}}\}.$$

Remark (Dependencies)*.*

- Reciprocal on $\mathbb{Q}$
- Reciprocal Lands in Pre $\mathbb{Q}$
- Reciprocal Respects Fraction Equivalence

Theorem

Theorem 2.2.36 (Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$)**.** *For every nonzero* $x \in \mathbb{Q}$*,*

$$x^{-1} \cdot x = 1_{\mathbb{Q}}.$$

Remark (Dependencies)*.*

- Reciprocal on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- One in $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined

Corollary

Corollary 2.2.37 (Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$)**.** *For every nonzero* $x \in \mathbb{Q}$*,*

$$x \cdot x^{-1} = 1_{\mathbb{Q}}.$$

Remark (Dependencies)*.*

- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.2.38 (Every Nonzero Rational Has a Multiplicative Inverse). *For every $x \in \mathbb{Q}$ with $x \neq 0_{\mathbb{Q}}$, there exists $y \in \mathbb{Q}$ such that*

$$x \cdot y = 1_{\mathbb{Q}} \qquad \text{and} \qquad y \cdot x = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$

Division on $\mathbb{Q}$

Definition

Definition 2.2.39 (Division on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$ with $y \neq 0_{\mathbb{Q}}$, define

$$x/y := x \cdot y^{-1}.$$

Remark (Dependencies).

- Multiplication on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$

Theorem

Theorem 2.2.40 (Division on $\mathbb{Q}$ Is Well-Defined). *Division determines a well-defined map*

$$\mathbb{Q} \times (\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}) \to \mathbb{Q}.$$

Remark (Dependencies).

- Division on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Reciprocal on $\mathbb{Q}$ Is Well-Defined

Distributivity on $\mathbb{Q}$-Classes

Theorem

Theorem 2.2.41 (Left Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined
- Multiplication Distributes over Addition on $\mathbb{Z}$

Corollary

Corollary 2.2.42 (Right Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Remark (Dependencies).

- Left Distributivity on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.2.43 (Multiplication Distributes over Addition on $\mathbb{Q}$). *Multiplication distributes over addition on both sides in $\mathbb{Q}$.*

Remark (Dependencies).

- Left Distributivity on $\mathbb{Q}$
- Right Distributivity on $\mathbb{Q}$

Class-Level Field Summary

Theorem

Theorem 2.2.44 ($\mathbb{Q}$ Is an Additive Abelian Group). *The structure $(\mathbb{Q}, +, 0_{\mathbb{Q}})$ is an abelian group.*

Remark (Dependencies).

- Two-Sided Inverse
- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Associative
- Addition on $\mathbb{Q}$ Is Commutative
- Zero Is an Additive Identity on $\mathbb{Q}$
- Every Rational Has an Additive Inverse

Theorem

Theorem 2.2.45 (Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group). *The structure* $(\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}, \cdot, 1_{\mathbb{Q}})$ *is an abelian group.*

Remark (Dependencies).

- Two-Sided Inverse
- Multiplication on $\mathbb{Q}$
- Nonzero Predicate Is Well-Defined on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$ Is Associative
- Multiplication on $\mathbb{Q}$ Is Commutative
- One Is a Multiplicative Identity on $\mathbb{Q}$
- Every Nonzero Rational Has a Multiplicative Inverse

Theorem

Theorem 2.2.46 ($\mathbb{Q}$ Is a Field). *The structure* $(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}})$ *is a field.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group
- Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group
- Multiplication Distributes over Addition on $\mathbb{Q}$
- Zero Is Not One in $\mathbb{Q}$

2.3 Field Structure of Q

Derived Field Laws on $\mathbb{Q}$

Corollary

Corollary 2.3.1 (Zero Product on $\mathbb{Q}$). *For all* $x, y \in \mathbb{Q}$,

$$x \cdot y = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}} \text{ or } y = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Zero Divisor

Corollary

Corollary 2.3.2 (Double Negation on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$-(-x) = x.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Two-Sided Inverse

Corollary

Corollary 2.3.3 (Negation of a Product on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \textit{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Two-Sided Inverse

Corollary

Corollary 2.3.4 (Product of Negatives on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Remark (Dependencies).

- Negation of a Product on $\mathbb{Q}$
- Double Negation on $\mathbb{Q}$

Corollary

Corollary 2.3.5 (Additive Cancellation on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x + z = y + z \implies x = y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group
- Two-Sided Inverse

Corollary

Corollary 2.3.6 (Multiplicative Cancellation on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z \neq 0_{\mathbb{Q}} \land x \cdot z = y \cdot z \implies x = y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Two-Sided Inverse

Corollary

Corollary 2.3.7 (Uniqueness of Additive Inverses on $\mathbb{Q}$). *Every additive inverse in $\mathbb{Q}$ is unique.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Additive Abelian Group
- Two-Sided Inverse

Corollary

Corollary 2.3.8 (Uniqueness of Multiplicative Inverses on $\mathbb{Q}$). *Every multiplicative inverse of a nonzero rational number is unique.*

Remark (Dependencies).

- Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group
- Two-Sided Inverse

Corollary

Corollary 2.3.9 (The Inverse of One Is One on $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$1_{\mathbb{Q}}^{-1} = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$
- One Is a Multiplicative Identity on $\mathbb{Q}$

Corollary

Corollary 2.3.10 (Inverse of a Product on $\mathbb{Q}$). *For all nonzero $x, y \in \mathbb{Q}$,*

$$(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Uniqueness of Multiplicative Inverses on $\mathbb{Q}$

Corollary

Corollary 2.3.11 (Inverse of an Inverse on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$(x^{-1})^{-1} = x.$$

Remark (Dependencies).

- Two-Sided Inverse

Integer Powers on $\mathbb{Q}$

Definition

Definition 2.3.12 (Integer Power on $\mathbb{Q}$)**.** Let $x \in \mathbb{Q}$. For $n \in \mathbb{W}$, define x^n by the usual recursive power operation. If $x \neq 0_{\mathbb{Q}}$ and $n \in \mathbb{Z}$ is negative, define

$$x^n := (x^{-1})^{-n}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- Every Nonzero Rational Has a Multiplicative Inverse
- Two-Sided Inverse

Corollary

Corollary 2.3.13 (Exponent Addition Law on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$x^{m+n} = x^m \cdot x^n.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$

Corollary

Corollary 2.3.14 (Power of a Power on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$(x^m)^n = x^{m \cdot n}.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$

Corollary

Corollary 2.3.15 (Power of a Product on $\mathbb{Q}$). *For all nonzero $x, y \in \mathbb{Q}$ and every $n \in \mathbb{Z}$,*

$$(x \cdot y)^n = x^n \cdot y^n.$$

Remark (Dependencies).

- Integer Power on $\mathbb{Q}$
- Integer Power on $\mathbb{Q}$
- Inverse of a Product on $\mathbb{Q}$

2.4 Order on $\mathbb{Q}$

Positivity on $\mathbb{Q}$

Definition

Definition 2.4.1 (Positive Rational). A rational number $[a, b] \in \mathbb{Q}$ is *positive* if

$$a \cdot b > 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- Rational Class Notation
- Multiplication on $\mathbb{Z}$
- Order on $\mathbb{Z}$

Lemma

Lemma 2.4.2 (Positivity Respects Fraction Equivalence on $\mathbb{Q}$). *If $[a, b] = [c, d]$, then*

$$a \cdot b > 0_{\mathbb{Z}} \iff c \cdot d > 0_{\mathbb{Z}}.$$

Remark (Dependencies).

- $\mathbb{Z}$ Is a Totally Ordered Set
- $\mathbb{Z}$ Has No Zero Divisors

Theorem

Theorem 2.4.3 (Positivity on $\mathbb{Q}$ Is Well-Defined). *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Q}$.*

Remark (Dependencies).

- Positive Rational
- Positivity Respects Fraction Equivalence on $\mathbb{Q}$

Definition

Definition 2.4.4 (Strict Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x < y \quad :\Longleftrightarrow \quad y - x \text{ is positive.}$$

Remark (Dependencies).

- Positive Rational
- Subtraction on $\mathbb{Q}$

Definition

Definition 2.4.5 (Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x \leq y \quad :\Longleftrightarrow \quad x < y \text{ or } x = y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$

Proposition

Proposition 2.4.6 (Product of Positives Is Positive on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \wedge 0_{\mathbb{Q}} < y \Longrightarrow 0_{\mathbb{Q}} < x \cdot y.$$

Remark (Dependencies).

- Positive Rational
- Multiplication on $\mathbb{Q}$
- $\mathbb{Z}$ Has No Zero Divisors

Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.7 (Sign Trichotomy on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$, exactly one of*

$$x < 0_{\mathbb{Q}}, \qquad x = 0_{\mathbb{Q}}, \qquad 0_{\mathbb{Q}} < x$$

holds.

Remark (Dependencies).

- Positive Rational
- Strict Order on $\mathbb{Q}$
- Zero in $\mathbb{Q}$

Theorem

Theorem 2.4.8 (Trichotomy on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Subtraction on $\mathbb{Q}$
- Sign Trichotomy on $\mathbb{Q}$

Total Order on $\mathbb{Q}$

Theorem

Theorem 2.4.9 (Strict Order on $\mathbb{Q}$ Is Irreflexive)**.** *For every $x \in \mathbb{Q}$, not $x < x$.*

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.10 (Strict Order on $\mathbb{Q}$ Is Asymmetric)**.** *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \neg(y < x).$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.11 (Strict Order on $\mathbb{Q}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \land y < z \implies x < z.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 2.4.12 (Order on $\mathbb{Q}$ Is Reflexive)**.** *For every $x \in \mathbb{Q}$, $x \leq x$.*

Remark (Dependencies).

- Order on $\mathbb{Q}$

Theorem

Theorem 2.4.13 (Order on $\mathbb{Q}$ Is Antisymmetric)**.** *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Strict Order on $\mathbb{Q}$ Is Asymmetric

Theorem

Theorem 2.4.14 (Order on $\mathbb{Q}$ Is Transitive)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Strict Order on $\mathbb{Q}$ Is Transitive

Theorem

Theorem 2.4.15 (Order on $\mathbb{Q}$ Is Total)**.** *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \text{ or } y \leq x.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.16 ($\mathbb{Q}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{Q}, \leq)$ is a totally ordered set.*

Remark (Dependencies).

- Ordered Set
- Total Order
- Order on $\mathbb{Q}$
- Order on $\mathbb{Q}$ Is Reflexive
- Order on $\mathbb{Q}$ Is Antisymmetric
- Order on $\mathbb{Q}$ Is Transitive
- Order on $\mathbb{Q}$ Is Total

Addition Order Compatibility on $\mathbb{Q}$

Theorem

Theorem 2.4.17 (Left Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \Longrightarrow z + x < z + y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Positive Rational

Corollary

Corollary 2.4.18 (Right Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \Longrightarrow x + z < y + z.$$

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Q}$
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.4.19 (Addition Preserves Strict Order on $\mathbb{Q}$). *Addition preserves strict order on both sides in $\mathbb{Q}$.*

Remark (Dependencies).

- Left Addition Preserves Strict Order on $\mathbb{Q}$
- Right Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 2.4.20 (Left Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \Longrightarrow z + x \leq z + y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Left Addition Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 2.4.21 (Right Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \Longrightarrow x + z \leq y + z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Right Addition Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 2.4.22 (Addition Preserves Order on $\mathbb{Q}$)**.** *Addition preserves non-strict order on both sides in* $\mathbb{Q}$*.*

Remark (Dependencies).

- Left Addition Preserves Order on $\mathbb{Q}$
- Right Addition Preserves Order on $\mathbb{Q}$

Corollary

Corollary 2.4.23 (Addition Reflects Order on $\mathbb{Q}$)**.** *Addition by a fixed rational reflects both strict and non-strict order.*

Remark (Dependencies).

- Addition Preserves Strict Order on $\mathbb{Q}$
- Addition Preserves Order on $\mathbb{Q}$
- Negation Gives a Left Additive Inverse on $\mathbb{Q}$

Multiplication Order Compatibility on $\mathbb{Q}$

Theorem

Theorem 2.4.24 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all* $x, y, z \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < z \wedge x < y \Longrightarrow z \cdot x < z \cdot y.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Product of Positives Is Positive on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$

Corollary

Corollary 2.4.25 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all* $x, y, z \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 2.4.26 (Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 2.4.27 (Left Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 2.4.28 (Right Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$

Theorem

Theorem 2.4.29 (Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves non-strict order on both sides.*

Remark (Dependencies).

- Left Multiplication by Positives Preserves Order on $\mathbb{Q}$
- Right Multiplication by Positives Preserves Order on $\mathbb{Q}$

Theorem

Theorem 2.4.30 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Remark (Dependencies).

- Strict Order on $\mathbb{Q}$
- Negation of a Product on $\mathbb{Q}$
- Positive Multiplication Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 2.4.31 (Multiplication by Negatives Reverses Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$

Reciprocal Order Laws on $\mathbb{Q}$

Theorem

Theorem 2.4.32 (Sign of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}, \qquad x < 0_{\mathbb{Q}} \Longrightarrow x^{-1} < 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- Product of Positives Is Positive on $\mathbb{Q}$
- Reciprocal Is a Left Multiplicative Inverse on $\mathbb{Q}$

Theorem

Theorem 2.4.33 (Reciprocal Reverses Order on Positives in $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x < y \Longrightarrow 0_{\mathbb{Q}} < y^{-1} < x^{-1}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- Sign of a Reciprocal on $\mathbb{Q}$
- Positive Multiplication Preserves Strict Order on $\mathbb{Q}$

Absolute Value on $\mathbb{Q}$

Definition

Definition 2.4.34 (Absolute Value on $\mathbb{Q}$)**.** For $x \in \mathbb{Q}$, define

$$|x| := \begin{cases} x, & 0_{\mathbb{Q}} \le x, \\ -x, & x < 0_{\mathbb{Q}}. \end{cases}$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Zero in $\mathbb{Q}$
- Negation on $\mathbb{Q}$

Theorem

Theorem 2.4.35 (Absolute Value Is Nonnegative on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \le |x|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.36 (Absolute Value Vanishes Only at Zero on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$|x| = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$
- Absolute Value Is Nonnegative on $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Theorem

Theorem 2.4.37 (Absolute Value Is Multiplicative on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$

- Positive Multiplication Preserves Order on $\mathbb{Q}$
- Product of Negatives on $\mathbb{Q}$

Theorem

Theorem 2.4.38 (Triangle Inequality on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x + y| \leq |x| + |y|.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$
- Addition on $\mathbb{Q}$
- Order on $\mathbb{Q}$
- $\mathbb{Q}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{Q}$

Theorem

Theorem 2.4.39 (Absolute Value of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$|x^{-1}| = |x|^{-1}.$$

Remark (Dependencies).

- Absolute Value on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$ Is Well-Defined
- Sign of a Reciprocal on $\mathbb{Q}$
- Absolute Value Is Multiplicative on $\mathbb{Q}$

2.5 Ordered Field Structure of Q

Ordered Field Instantiation

Theorem

Theorem 2.5.1 ($\mathbb{Q}$ Is an Ordered Field)**.** *The structure*

$$(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}}, \leq)$$

is an ordered field.

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field

- $\mathbb{Q}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{Q}$
- Multiplication by Positives Preserves Order on $\mathbb{Q}$

Corollary

Corollary 2.5.2 (Ordered Field Laws Hold on $\mathbb{Q}$)**.** *Every theorem that depends only on the ordered-field axioms applies to $\mathbb{Q}$ with its constructed operations and order.*

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Strict Total Order

Standard Ordered-Field Consequences

Corollary

Corollary 2.5.3 (Squares Are Nonnegative on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq x^2.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Product of Positives Is Positive on $\mathbb{Q}$

Corollary

Corollary 2.5.4 (One Is Positive in $\mathbb{Q}$)**.** *In $\mathbb{Q}$,*

$$0_{\mathbb{Q}} < 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- One in $\mathbb{Q}$

Corollary

Corollary 2.5.5 (Natural Numbers Are Positive in $\mathbb{Q}$)**.** *Every natural number embedded in $\mathbb{Q}$ is positive.*

Remark (Dependencies).

- One Is Positive in $\mathbb{Q}$

Corollary

Corollary 2.5.6 (Positive Rationals Have Positive Inverses)**.** *For every* $x \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$

Corollary

Corollary 2.5.7 (Fraction Comparison in $\mathbb{Q}$)**.** *For rational classes with positive denominator product,*

$$[a, b] < [c, d] \quad \Longleftrightarrow \quad a \cdot d < c \cdot b.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Product of Positives Is Positive on $\mathbb{Q}$
- One in $\mathbb{Q}$
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$
- Positive Multiplication Preserves Order on $\mathbb{Q}$
- Reciprocal Reverses Order on Positives in $\mathbb{Q}$

2.6 Arbitrary Closeness in $\mathbb{Q}$

Density and the Archimedean Property

Definition

Definition 2.6.1 (Two in $\mathbb{Q}$)**.** Define

$$2_{\mathbb{Q}} := 1_{\mathbb{Q}} + 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- One in $\mathbb{Q}$
- Addition on $\mathbb{Q}$

Lemma

Lemma 2.6.2 (Two Is Nonzero in $\mathbb{Q}$)**.** *In* $\mathbb{Q}$*,*

$$2_{\mathbb{Q}} \neq 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- One Is Positive in $\mathbb{Q}$
- Trichotomy on $\mathbb{Q}$

Lemma

Lemma 2.6.3 (Midpoint Lies Strictly Between Rationals)**.** *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies x < (x+y) \cdot 2_{\mathbb{Q}}^{-1} < y.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Two Is Nonzero in $\mathbb{Q}$

Theorem

Theorem 2.6.4 (Density of $\mathbb{Q}$ in Itself)**.** *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \exists r \in \mathbb{Q}\ (x < r < y).$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Midpoint Lies Strictly Between Rationals

Theorem

Theorem 2.6.5 (Archimedean Property of $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies \exists n \in \mathbb{N}\ (y < n \cdot x).$$

Remark (Dependencies).

- Well-Ordering Principle
- Division Algorithm on $\mathbb{Z}$
- $\mathbb{Z}$ Is a Totally Ordered Set

From Discreteness to Arbitrary Closeness

Remark 2.6.6 (Discrete Systems Before $\mathbb{Q}$). The systems $\mathbb{N}$, $\mathbb{W}$, and $\mathbb{Z}$ are discrete in the order sense: consecutive integer points have no same-system point between them. The rational numbers are the first number system in this construction chain where intervals can be subdivided internally without leaving the system.

Definition

Definition 2.6.7 (Arbitrarily Close Distinct Points)**.** A number system has arbitrarily close distinct points if for every point x and every positive tolerance ε, there is a point $y \neq x$ such that

$$|x - y| < \varepsilon.$$

Remark (Dependencies).

- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$

Theorem

Theorem 2.6.8 ($\mathbb{Q}$ Has Arbitrarily Close Distinct Points)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y \in \mathbb{Q}\ (y \neq x \text{ and } |x - y| < \varepsilon).$$

Remark (Dependencies).

- Reciprocals Become Arbitrarily Small in $\mathbb{Q}$
- Addition Preserves Strict Order on $\mathbb{Q}$

Corollary

Corollary 2.6.9 ($\mathbb{Q}$ Has No Adjacent Points)**.** *There do not exist rational numbers $x < y$ with no rational number strictly between them.*

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself

Proposition

Proposition 2.6.10 ($\mathbb{Q}$ Has No Least Positive Rational)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies \exists y \in \mathbb{Q}\ (0_{\mathbb{Q}} < y < x).$$

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself

Corollary

Corollary 2.6.11 (Reciprocals Become Arbitrarily Small in $\mathbb{Q}$)**.** *For every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists n \in \mathbb{N}\ \left(0_{\mathbb{Q}} < n^{-1} < \varepsilon\right).$$

Remark (Dependencies).

- Archimedean Property of $\mathbb{Q}$
- Sign of a Reciprocal on $\mathbb{Q}$

Limit-Like Behavior Before Limits

Proposition

Proposition 2.6.12 (One-Sided Rational Approximation). *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y, z \in \mathbb{Q}\ (x - \varepsilon < y < x < z < x + \varepsilon).$$

Remark (Dependencies).

- $\mathbb{Q}$ Has Arbitrarily Close Distinct Points
- Density of $\mathbb{Q}$ in Itself

Remark 2.6.13 (Refinement and Bounds). The Archimedean property supplies arbitrarily small rational increments, so a rational point can be approached from one side at any prescribed tolerance. Mediants supply a complementary order-refinement mechanism: once two rational bounds are known, their mediant is a new rational bound strictly between them. The following approximation topics make these two mechanisms explicit through dyadic grids, mediant refinement, Stern–Brocot paths, and rational Cauchy sequences.

2.7 Rational Gaps

No Rational Square Root of Two

Theorem

Theorem 2.7.1 (No Rational Square Root of Two). *There is no rational number $q \in \mathbb{Q}$ such that*

$$q \cdot q = 2_{\mathbb{Q}}.$$

Remark (Dependencies).

- $\mathbb{Q}$ Is a Field
- $\mathbb{Z}$ Is an Integral Domain
- TODO: formalize the parity or infinite-descent facts in $\mathbb{Z}$.

The Square-Two Gap

Definition

Definition 2.7.2 (The Gap Set in $\mathbb{Q}$). Define

$$S := \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \text{ and } q \cdot q < 2_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rationals
- Order on $\mathbb{Q}$

- Multiplication on $\mathbb{Q}$
- Zero in $\mathbb{Q}$

Proposition

Proposition 2.7.3 (The Gap Set Is Bounded Above in $\mathbb{Q}$)**.** *The set S is nonempty and bounded above in $\mathbb{Q}$.*

Remark (Dependencies)*.*

- $\mathbb{Q}$ Is an Ordered Field
- Natural Numbers Are Positive in $\mathbb{Q}$

Theorem

Theorem 2.7.4 (The Square-Two Gap Has No Rational Supremum)**.** *The set S has no least upper bound in $\mathbb{Q}$.*

Remark (Dependencies)*.*

- No Rational Square Root of Two
- Density of $\mathbb{Q}$ in Itself
- $\mathbb{Q}$ Is an Ordered Field

Corollary

Corollary 2.7.5 ($\mathbb{Q}$ Is Order-Incomplete)**.** *The ordered field $\mathbb{Q}$ is not order-complete: there exists a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.*

Remark (Dependencies)*.*

- The Gap Set Is Bounded Above in $\mathbb{Q}$
- The Square-Two Gap Has No Rational Supremum

Cauchy Sequences in $\mathbb{Q}$

Definition

Definition 2.7.6 (Cauchy Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is Cauchy if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall m, n \geq N\ |x_m - x_n| < \varepsilon)\,.$$

Remark (Dependencies)*.*

- Rationals
- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$

- Subtraction on $\mathbb{Q}$

Definition

Definition 2.7.7 (Null Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is null if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \implies \exists N \in \mathbb{N}\, \forall n \geq N\ |x_n| < \varepsilon)\,.$$

Remark (Dependencies).

- Rationals
- Order on $\mathbb{Q}$
- Absolute Value on $\mathbb{Q}$

Lemma

Lemma 2.7.8 (Rational Limits Are Unique)**.** *A convergent sequence in $\mathbb{Q}$ has at most one rational limit.*

Remark (Dependencies).

- Triangle Inequality on $\mathbb{Q}$
- $\mathbb{Q}$ Is an Ordered Field

2.8 Embedding $\mathbb{Z}$ into $\mathbb{Q}$

The Integer Embedding

Definition

Definition 2.8.1 (Embedding of $\mathbb{Z}$ into $\mathbb{Q}$)**.** Define

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(n) := [n, 1_{\mathbb{Z}}].$$

Remark (Dependencies).

- Rational Numbers
- One in $\mathbb{Z}$
- Zero Is Not One in $\mathbb{Z}$

Theorem

Theorem 2.8.2 (Embedding Preserves Zero)**.** *The embedding sends integer zero to rational zero:*

$$\iota(0_{\mathbb{Z}}) = 0_{\mathbb{Q}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Z}$ into $\mathbb{Q}$

- Zero in $\mathbb{Z}$
- Zero in $\mathbb{Q}$

Theorem

Theorem 2.8.3 (Embedding Preserves One). *The embedding sends integer one to rational one:*

$$\iota(1_{\mathbb{Z}}) = 1_{\mathbb{Q}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Z}$ into $\mathbb{Q}$
- One in $\mathbb{Z}$
- One in $\mathbb{Q}$

Theorem

Theorem 2.8.4 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m) = \iota(n) \implies m = n.$$

Remark (Dependencies).

- Equality Criterion for Rational Classes
- One Is a Multiplicative Identity on $\mathbb{Z}$

Preservation of Algebra and Order

Theorem

Theorem 2.8.5 (Embedding Preserves Addition). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m + n) = \iota(m) + \iota(n).$$

Remark (Dependencies).

- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Addition on $\mathbb{Z}$
- Addition on $\mathbb{Q}$
- Addition on $\mathbb{Z}$ Is Well-Defined
- Addition on $\mathbb{Q}$ Is Well-Defined

Theorem

Theorem 2.8.6 (Embedding Preserves Multiplication). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m \cdot n) = \iota(m) \cdot \iota(n).$$

Remark (Dependencies).

- Embedding $\mathbb{Z}$ into $\mathbb{Q}$
- Multiplication on $\mathbb{Z}$
- Multiplication on $\mathbb{Q}$
- Multiplication on $\mathbb{Z}$ Is Well-Defined
- Multiplication on $\mathbb{Q}$ Is Well-Defined

Theorem

Theorem 2.8.7 (Embedding Preserves Order)**.** *For all $m, n \in \mathbb{Z}$,*

$$m \leq n \iff \iota(m) \leq \iota(n).$$

Remark (Dependencies).

- Addition on $\mathbb{Q}$-Classes
- Multiplication on $\mathbb{Q}$-Classes
- Order on $\mathbb{Q}$
- $\mathbb{Z}$ Is a Totally Ordered Set

Theorem

Theorem 2.8.8 ($\mathbb{Z}$ Embeds into $\mathbb{Q}$)**.** *The map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an injective order-preserving ring homomorphism.*

Remark (Dependencies).

- Embedding Preserves Zero
- Embedding Preserves One
- Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective
- Embedding Preserves Addition
- Embedding Preserves Multiplication
- Embedding Preserves Order

Field of Fractions

Theorem

Theorem 2.8.9 ($\mathbb{Q}$ Is the Field of Fractions of $\mathbb{Z}$)**.** *For every $x \in \mathbb{Q}$, there exist $m, n \in \mathbb{Z}$ with $n \neq 0_{\mathbb{Z}}$ such that*

$$x = \iota(m) \cdot \iota(n)^{-1}.$$

Remark (Dependencies).

- Rational Class Notation
- Every Nonzero Rational Has a Multiplicative Inverse
- Vanishing Criterion

Corollary

Corollary 2.8.10 (The Embedding of $\mathbb{Z}$ into $\mathbb{Q}$ Is Not Surjective)**.** *The image of $\iota : \mathbb{Z} \to \mathbb{Q}$ is a proper subset of $\mathbb{Q}$.*

Remark (Dependencies).

- Density of $\mathbb{Q}$ in Itself
- Embedding Preserves Order

2.9 Transition to $\mathbb{R}$

Why the Real Numbers Are Introduced

Remark 2.9.1 (Why the Real Numbers Are Introduced)*.* The rational numbers form an ordered field, are dense in themselves, and satisfy the Archimedean property. These facts make $\mathbb{Q}$ a rich arithmetic and order structure, but they do not make it complete.

The square-two gap shows the defect precisely: there is a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$. The real numbers are introduced as a completion of $\mathbb{Q}$, so that bounded-above sets acquire suprema and Cauchy sequences acquire limits.

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Density of $\mathbb{Q}$ in Itself
- Archimedean Property of $\mathbb{Q}$
- $\mathbb{Q}$ Is Order-Incomplete
- Cauchy Sequence in $\mathbb{Q}$

2.10 Canonical Rational Number Statement Plan

Architecture Rule for the Rational Numbers

The rational-number chapter is organized around a local-versus-generic separation. Local results open the quotient construction box: prefractions, fraction equivalence, well-defined operations, field axioms, order, density, gaps, and the embedding of $\mathbb{Z}$ into $\mathbb{Q}$. Generic field and ring consequences are not re-proved here. They are cited as corollaries from the general algebraic infrastructure developed earlier.

Remark (Partiality of reciprocal)*.* The new structural issue relative to $\mathbb{Z}$ is that reciprocal is a partial operation. It is defined only on nonzero rational classes, so the nonzero predicate must be proved well-defined before reciprocal is introduced.

Construction of $\mathbb{Q}$

- Def `def:nonzero-integers`: $\mathbb{Z}^* := \mathbb{Z} \setminus \{0\}$.
- Def `def:prefraction`: a prefraction is $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as a/b.
- Def `def:prefraction-set`: $\mathrm{PreQ} := \mathbb{Z} \times \mathbb{Z}^*$.
- Def `def:fraction-equivalence`: $(a, b) \sim_Q (c, d)$ iff $ad = bc$.
- Lem `lem:fraction-equivalence-reflexive`: $(a, b) \sim_Q (a, b)$.
- Lem `lem:fraction-equivalence-symmetric`: $(a, b) \sim_Q (c, d) \Rightarrow (c, d) \sim_Q (a, b)$.
- Lem `lem:fraction-equivalence-transitive`: fraction equivalence is transitive; depends on nonzero multiplicative cancellation in $\mathbb{Z}$.
- Thm `thm:fraction-equivalence-relation`: $\sim_Q$ is an equivalence relation on PreQ.
- Def `def:rationals`: $\mathbb{Q} := \mathrm{PreQ} / \sim_Q$.
- Def `def:rational-class-notation`: $[a, b]$ denotes the $\sim_Q$-class of (a, b) with $b \neq 0$.
- Def `def:zero-in-q`: $0_Q := [0, 1]$.
- Def `def:one-in-q`: $1_Q := [1, 1]$.
- Lem `lem:vanishing-criterion-q`: $[a, b] = 0_Q$ iff $a = 0$.
- Thm `thm:zero-is-a-rational`: $0_Q \in \mathbb{Q}$.
- Thm `thm:one-is-a-rational`: $1_Q \in \mathbb{Q}$.
- Thm `thm:zero-not-one-in-q`: $0_Q \neq 1_Q$; depends on $0_Z \neq 1_Z$.

Operations on Rational Classes

- Def `def:addition-on-q-classes`: $[a, b] + [c, d] := [ad + bc, bd]$.
- Lem `lem:addition-denominator-nonzero-q`: $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$.
- Lem `lem:addition-on-q-respects-left-equivalence`: addition respects fraction equivalence in the left argument.
- Lem `lem:addition-on-q-respects-right-equivalence`: addition respects fraction equivalence in the right argument.
- Lem `lem:addition-on-q-respects-equivalence`: addition respects fraction equivalence in both arguments.

- Thm `thm:addition-on-q-well-defined`: addition is a well-defined operation $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.
- Thm `thm:addition-on-q-associative`: $(x + y) + z = x + (y + z)$.
- Thm `thm:addition-on-q-commutative`: $x + y = y + x$.
- Thm `thm:zero-left-additive-identity-on-q`: $0_Q + x = x$.
- Cor `cor:zero-right-additive-identity-on-q`: $x + 0_Q = x$.
- Def `def:negation-on-q-classes`: $-[a, b] := [-a, b]$.
- Lem `lem:negation-on-q-respects-equivalence`: negation respects fraction equivalence.
- Thm `thm:negation-on-q-well-defined`: negation is well-defined on $\mathbb{Q}$.
- Thm `thm:negation-left-additive-inverse-on-q`: $(-x) + x = 0_Q$.
- Cor `cor:negation-right-additive-inverse-on-q`: $x + (-x) = 0_Q$.
- Thm `thm:every-rational-has-additive-inverse`: every rational has a two-sided additive inverse.
- Def `def:subtraction-on-q`: $x - y := x + (-y)$.
- Thm `thm:subtraction-on-q-well-defined`: subtraction is well-defined.
- Def `def:multiplication-on-q-classes`: $[a, b] \cdot [c, d] := [ac, bd]$.
- Lem `lem:multiplication-denominator-nonzero-q`: $b \neq 0$ and $d \neq 0$ imply $bd \neq 0$.
- Lem `lem:multiplication-on-q-respects-left-equivalence`: multiplication respects equivalence in the left argument.
- Lem `lem:multiplication-on-q-respects-right-equivalence`: multiplication respects equivalence in the right argument.
- Lem `lem:multiplication-on-q-respects-equivalence`: multiplication respects equivalence in both arguments.
- Thm `thm:multiplication-on-q-well-defined`: multiplication is well-defined on $\mathbb{Q}$.
- Thm `thm:multiplication-on-q-associative`: $(xy)z = x(yz)$.
- Thm `thm:multiplication-on-q-commutative`: $xy = yx$.
- Thm `thm:one-left-multiplicative-identity-on-q`: $1_Q x = x$.
- Cor `cor:one-right-multiplicative-identity-on-q`: $x 1_Q = x$.

- Lem `lem:nonzero-predicate-well-defined-q`: $[a, b] \neq 0_Q$ iff $a \neq 0$.
- Def `def:reciprocal-on-q-classes`: for $a \neq 0$, $[a, b]^{-1} := [b, a]$.
- Lem `lem:reciprocal-denominator-nonzero-q`: $a \neq 0$ puts (b, a) in PreQ.
- Lem `lem:reciprocal-on-q-respects-equivalence`: reciprocal respects equivalence on nonzero classes.
- Thm `thm:reciprocal-on-q-well-defined`: reciprocal is a well-defined map $\mathbb{Q} \setminus \{0_Q\} \to \mathbb{Q} \setminus \{0_Q\}$.
- Thm `thm:reciprocal-left-multiplicative-inverse-on-q`: $x \neq 0_Q \Rightarrow x^{-1}x = 1_Q$.
- Cor `cor:reciprocal-right-multiplicative-inverse-on-q`: $x \neq 0_Q \Rightarrow xx^{-1} = 1_Q$.
- Thm `thm:every-nonzero-rational-has-multiplicative-inverse`: every nonzero rational has a two-sided multiplicative inverse.
- Def `def:division-on-q`: for $y \neq 0_Q$, $x/y := x \cdot y^{-1}$.
- Thm `thm:division-on-q-well-defined`: division is well-defined on $\mathbb{Q} \times (\mathbb{Q} \setminus \{0_Q\})$.
- Thm `thm:left-distributivity-on-q`: $x(y + z) = xy + xz$.
- Cor `cor:right-distributivity-on-q`: $(y + z)x = yx + zx$.

Field Structure and Generic Field Laws

- Thm `thm:q-additive-abelian-group`: $(\mathbb{Q}, +, 0_Q)$ is an abelian group.
- Thm `thm:q-nonzero-multiplicative-abelian-group`: $(\mathbb{Q} \setminus \{0_Q\}, \cdot, 1_Q)$ is an abelian group.
- Thm `thm:q-is-a-field`: $\mathbb{Q}$ is a field.
- Cor `cor:zero-product-on-q`: $xy = 0_Q$ iff $x = 0_Q$ or $y = 0_Q$; cites the generic field zero-product theorem.
- Cor `cor:double-negation-on-q`: $-(-x) = x$; cites the generic ring result.
- Cor `cor:negation-of-product-on-q`: $(-x)y = -(xy) = x(-y)$; cites the generic ring result.
- Cor `cor:product-of-negatives-on-q`: $(-x)(-y) = xy$; cites the generic ring result.
- Cor `cor:additive-cancellation-on-q`: $x + z = y + z \Rightarrow x = y$; cites generic group cancellation.

- Cor `cor:multiplicative-cancellation-on-q`: $z \neq 0_Q$ and $xz = yz$ imply $x = y$; cites generic field cancellation.
- Cor `cor:uniqueness-additive-inverse-on-q`: additive inverses are unique.
- Cor `cor:uniqueness-multiplicative-inverse-on-q`: multiplicative inverses of nonzero elements are unique.
- Cor `cor:inverse-of-one-is-one-on-q`: $1_Q^{-1} = 1_Q$.
- Cor `cor:inverse-of-product-on-q`: if $x, y \neq 0$, then $(xy)^{-1} = x^{-1}y^{-1}$.
- Cor `cor:inverse-of-inverse-on-q`: if $x \neq 0$, then $(x^{-1})^{-1} = x$.
- Def `def:integer-power-on-q`: x^n for $n \in \mathbb{Z}$, with $x \neq 0_Q$ when $n < 0$.
- Cor `cor:exponent-addition-on-q`: $x^{m+n} = x^m x^n$.
- Cor `cor:power-of-power-on-q`: $(x^m)^n = x^{mn}$.
- Cor `cor:power-of-product-on-q`: $(xy)^n = x^n y^n$.

Order on $\mathbb{Q}$

- Def `def:positive-rational`: $[a, b]$ is positive iff $ab > 0$ in $\mathbb{Z}$.
- Lem `lem:positivity-respects-equivalence-q`: positivity respects fraction equivalence.
- Thm `thm:positivity-on-q-well-defined`: positivity is well-defined on $\mathbb{Q}$.
- Def `def:strict-order-on-q`: $x < y$ iff $y - x$ is positive.
- Def `def:order-on-q`: $x \leq y$ iff $x < y$ or $x = y$.
- Prop `prop:product-of-positives-is-positive-q`: positive rationals have positive product.
- Thm `thm:sign-trichotomy-on-q`: exactly one of $x < 0_Q$, $x = 0_Q$, $0_Q < x$ holds.
- Thm `thm:trichotomy-on-q`: exactly one of $x < y$, $x = y$, $y < x$ holds.
- Thm `thm:strict-order-on-q-irreflexive`: not $x < x$.
- Thm `thm:strict-order-on-q-asymmetric`: $x < y \Rightarrow \neg(y < x)$.
- Thm `thm:strict-order-on-q-transitive`: $x < y$ and $y < z$ imply $x < z$.
- Thm `thm:order-on-q-reflexive`: $x \leq x$.
- Thm `thm:order-on-q-antisymmetric`: $x \leq y$ and $y \leq x$ imply $x = y$.
- Thm `thm:order-on-q-transitive`: $x \leq y$ and $y \leq z$ imply $x \leq z$.

- Thm `thm:order-on-q-total`: $x \leq y$ or $y \leq x$.
- Thm `thm:q-totally-ordered-set`: $(\mathbb{Q}, \leq)$ is a total order.
- Thm `thm:left-addition-preserves-strict-order-on-q`: $x < y \Rightarrow z + x < z + y$.
- Cor `cor:right-addition-preserves-strict-order-on-q`: $x < y \Rightarrow x + z < y + z$.
- Thm `thm:left-addition-preserves-order-on-q`: $x \leq y \Rightarrow z + x \leq z + y$.
- Cor `cor:right-addition-preserves-order-on-q`: $x \leq y \Rightarrow x + z \leq y + z$.
- Cor `cor:addition-reflects-order-on-q`: addition by a fixed rational reflects strict and non-strict order.
- Thm `thm:left-positive-multiplication-preserves-strict-order-on-q`: positive left multiplication preserves strict order.
- Cor `cor:right-positive-multiplication-preserves-strict-order-on-q`: positive right multiplication preserves strict order.
- Thm `thm:positive-multiplication-preserves-order-on-q`: positive multiplication preserves non-strict order.
- Thm `thm:negative-multiplication-reverses-strict-order-on-q`: negative multiplication reverses strict order.
- Thm `thm:reciprocal-reverses-order-on-positives-q`: $0_Q < x < y \Rightarrow 0_Q < y^{-1} < x^{-1}$.
- Thm `thm:sign-of-reciprocal-q`: reciprocals preserve positive and negative sign.
- Def `def:absolute-value-on-q`: $|x| := x$ if $0_Q \leq x$, and $|x| := -x$ otherwise.
- Thm `thm:absolute-value-nonnegative-on-q`: $0_Q \leq |x|$.
- Thm `thm:absolute-value-zero-iff-on-q`: $|x| = 0_Q$ iff $x = 0_Q$.
- Thm `thm:absolute-value-multiplicative-on-q`: $|xy| = |x||y|$.
- Thm `thm:triangle-inequality-on-q`: $|x + y| \leq |x| + |y|$.
- Thm `thm:absolute-value-of-reciprocal-q`: $x \neq 0_Q \Rightarrow |x^{-1}| = |x|^{-1}$.
- Thm `thm:q-is-an-ordered-field`: $\mathbb{Q}$ is an ordered field.

Density, Gaps, and Embedding

- Lem `lem:midpoint-between-q`: $x < y \Rightarrow x < (x + y)2_Q^{-1} < y$.
- Thm `thm:density-of-q`: between any two rationals lies a rational.

- Prop `prop:no-least-positive-rational`: every positive rational has a smaller positive rational.

- Thm `thm:archimedean-property-of-q`: for $0_Q < x$, some $n \in \mathbb{N}$ has $y < nx$.

- Cor `cor:reciprocals-arbitrarily-small-q`: reciprocals of naturals become arbitrarily small.

- Thm `thm:no-rational-square-root-of-two`: no rational square is 2_Q.

- Def `def:gap-set-q`: $S := \{q \in \mathbb{Q} : 0_Q < q \wedge q^2 < 2_Q\}$.

- Prop `prop:gap-set-bounded-above-q`: the gap set is nonempty and bounded above.

- Thm `thm:gap-set-no-rational-supremum-q`: the gap set has no rational supremum.

- Cor `cor:q-order-incomplete`: $\mathbb{Q}$ is not order-complete.

- Def `def:cauchy-sequence-in-q`: rational Cauchy sequence.

- Def `def:null-sequence-in-q`: rational null sequence.

- Lem `lem:rational-limit-unique`: a convergent rational sequence has at most one rational limit.

- Def `def:embedding-z-into-q`: $\iota : \mathbb{Z} \to \mathbb{Q}$, $\iota(n) := [n, 1]$.

- Thm `thm:embedding-z-into-q-preserves-zero`: $\iota(0_Z) = 0_Q$.

- Thm `thm:embedding-z-into-q-preserves-one`: $\iota(1_Z) = 1_Q$.

- Thm `thm:embedding-z-into-q-injective`: $\iota(m) = \iota(n) \Rightarrow m = n$.

- Thm `thm:embedding-z-into-q-preserves-addition`: $\iota(m + n) = \iota(m) + \iota(n)$.

- Thm `thm:embedding-z-into-q-preserves-multiplication`: $\iota(mn) = \iota(m)\iota(n)$.

- Thm `thm:embedding-z-into-q-preserves-order`: $m \leq n$ iff $\iota(m) \leq \iota(n)$.

- Thm `thm:z-embeds-into-q`: ι is an injective order-preserving ring homomorphism.

- Thm `thm:q-is-field-of-fractions-of-z`: every rational is a quotient of integer images.

- Cor `cor:embedding-z-into-q-not-surjective`: $\iota(\mathbb{Z}) \subsetneq \mathbb{Q}$.

- Rem `rem:why-real-numbers-are-introduced`: $\mathbb{Q}$ is an ordered field but order-incomplete, motivating $\mathbb{R}$.

Chapter 3

Real Numbers ($\mathbb{R}$)

reals

Rational Numbers $\to$ **Real Numbers** $\to$ Structure of the Real Line

3.1 Construction of $\mathbb{R}$

Dedekind Cuts

Definition

Definition 3.1.1 (Dedekind Cut)**.** A *Dedekind cut* is a set $\alpha \subseteq \mathbb{Q}$ satisfying:

$$\alpha \neq \emptyset, \qquad \alpha \neq \mathbb{Q},$$

$$p \in \alpha \land q < p \implies q \in \alpha,$$

and

$$p \in \alpha \implies \exists r \in \alpha \, (p < r).$$

Remark (Dependencies).

- Density of $\mathbb{Q}$

Definition

Definition 3.1.2 (The Real Numbers)**.** The real numbers are the set

$$\mathbb{R} := \{\alpha : \alpha \text{ is a Dedekind cut in } \mathbb{Q}\}.$$

Remark (Dependencies).

- Dedekind Cut

Rational Cuts

Definition

Definition 3.1.3 (Rational Cut)**.** For $q \in \mathbb{Q}$, define the rational cut

$$q^* := \{r \in \mathbb{Q} : r < q\}.$$

Remark (Dependencies).

- Rational Numbers
- Order on $\mathbb{Q}$

Lemma

Lemma 3.1.4 (Rational Cuts Are Cuts)**.** *For every $q \in \mathbb{Q}$, the set q^* is a Dedekind cut.*

Remark (Dependencies).

- Density of $\mathbb{Q}$
- No Least Positive Rational

Zero and One in $\mathbb{R}$

Definition

Definition 3.1.5 (Zero in $\mathbb{R}$)**.** Define

$$0_{\mathbb{R}} := (0_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Cut
- Zero in $\mathbb{Q}$

Definition

Definition 3.1.6 (One in $\mathbb{R}$)**.** Define

$$1_{\mathbb{R}} := (1_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 1_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Cut
- One in $\mathbb{Q}$

Theorem

Theorem 3.1.7 (Zero and One Are Reals)**.** *The cuts $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$ are elements of $\mathbb{R}$.*

Remark (Dependencies).

- The Real Numbers
- Zero in $\mathbb{R}$
- One in $\mathbb{R}$

- Rational Cuts Are Cuts

Theorem

Theorem 3.1.8 (Zero Is Not One in $\mathbb{R}$). *In* $\mathbb{R}$,

$$0_{\mathbb{R}} \neq 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Zero in $\mathbb{R}$
- One in $\mathbb{R}$
- One Is Positive in $\mathbb{Q}$

3.2 Order on $\mathbb{R}$

Order by Inclusion

Definition

Definition 3.2.1 (Order on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \leq \beta \quad :\Longleftrightarrow \quad \alpha \subseteq \beta.$$

Remark (Dependencies).

- The Real Numbers
- Subset

Definition

Definition 3.2.2 (Strict Order on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \subsetneq \beta.$$

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Proper Subset

Theorem

Theorem 3.2.3 (Order on $\mathbb{R}$ Is Reflexive). *For every* $\alpha \in \mathbb{R}$, $\alpha \leq \alpha$.

Remark (Dependencies).

- Order on $\mathbb{R}$
- Subset

Theorem

Theorem 3.2.4 (Order on $\mathbb{R}$ Is Antisymmetric)**.** *For all* $\alpha, \beta \in \mathbb{R}$*,*

$$\alpha \leq \beta \wedge \beta \leq \alpha \implies \alpha = \beta.$$

Remark (Dependencies)*.*

- Order on $\mathbb{R}$
- Set Equality
- Subset

Theorem

Theorem 3.2.5 (Order on $\mathbb{R}$ Is Transitive)**.** *For all* $\alpha, \beta, \gamma \in \mathbb{R}$*,*

$$\alpha \leq \beta \wedge \beta \leq \gamma \implies \alpha \leq \gamma.$$

Remark (Dependencies)*.*

- Order on $\mathbb{R}$
- Subset

Theorem

Theorem 3.2.6 (Comparability of Cuts)**.** *For all cuts* $\alpha, \beta \in \mathbb{R}$*,*

$$\alpha \subseteq \beta \vee \beta \subseteq \alpha.$$

Remark (Dependencies)*.*

- Dedekind Cut
- $\mathbb{Q}$ Is a Totally Ordered Set

Theorem

Theorem 3.2.7 (Order on $\mathbb{R}$ Is Total)**.** *For all* $\alpha, \beta \in \mathbb{R}$*,*

$$\alpha \leq \beta \vee \beta \leq \alpha.$$

Remark (Dependencies)*.*

- Order on $\mathbb{R}$
- Comparability of Cuts

Theorem

Theorem 3.2.8 (Trichotomy on $\mathbb{R}$)**.** *For all* $\alpha, \beta \in \mathbb{R}$*, exactly one of*

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha$$

holds.

Remark (Dependencies)*.*

- Strict Order on $\mathbb{R}$
- Order on $\mathbb{R}$ Is Antisymmetric
- Order on $\mathbb{R}$ Is Total

Theorem

Theorem 3.2.9 ($\mathbb{R}$ Is a Totally Ordered Set)**.** *The ordered pair* $(\mathbb{R}, \leq)$ *is a total order.*

Remark (Dependencies).

- Ordered Set
- Total Order
- Order on $\mathbb{R}$
- Order on $\mathbb{R}$ Is Reflexive
- Order on $\mathbb{R}$ Is Antisymmetric
- Order on $\mathbb{R}$ Is Transitive
- Order on $\mathbb{R}$ Is Total

3.3 Completeness of $\mathbb{R}$

Definition

Definition 3.3.1 (Upper Bound in $\mathbb{R}$)**.** Let $S \subseteq \mathbb{R}$. A cut $\gamma \in \mathbb{R}$ is an upper bound of S if

$$\forall \alpha \in S \, (\alpha \leq \gamma).$$

The set S is bounded above if it has an upper bound in $\mathbb{R}$.

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Subset

Definition

Definition 3.3.2 (Supremum in $\mathbb{R}$)**.** For $S \subseteq \mathbb{R}$, a cut σ is the supremum of S if σ is an upper bound of S and every upper bound γ of S satisfies $\sigma \leq \gamma$.

Remark (Dependencies).

- Upper Bound in $\mathbb{R}$
- Order on $\mathbb{R}$

Definition

Definition 3.3.3 (Least-Upper-Bound Property on $\mathbb{R}$)**.** The ordered set $\mathbb{R}$ has the *least-upper-bound property* if every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Upper Bound in $\mathbb{R}$
- Supremum in $\mathbb{R}$

Lemma

Lemma 3.3.4 (Union of a Bounded Nonempty Family of Cuts Is a Cut)**.** *If $S \subseteq \mathbb{R}$ is nonempty and bounded above, then*

$$\bigcup_{\alpha \in S} \alpha$$

is a Dedekind cut.

Remark (Dependencies).

- Dedekind Cut
- Upper Bound in $\mathbb{R}$
- Union

Theorem

Theorem 3.3.5 (Least-Upper-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound, namely*

$$\sup S = \bigcup_{\alpha \in S} \alpha.$$

Remark (Dependencies).

- Least-Upper-Bound Property on $\mathbb{R}$
- Union of a Bounded Nonempty Family of Cuts Is a Cut
- Supremum in $\mathbb{R}$

Corollary

Corollary 3.3.6 (Greatest-Lower-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound.*

Remark (Dependencies).

- Least-Upper-Bound Property

3.4 Addition on $\mathbb{R}$

Definition

Definition 3.4.1 (Addition of Cuts)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta := \{p + q : p \in \alpha,\ q \in \beta\}.$$

Remark (Dependencies).

- The Real Numbers
- Addition on $\mathbb{Q}$

Lemma

Lemma 3.4.2 (Sum of Cuts Is a Cut)**.** *If $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is a Dedekind cut.*

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 3.4.3 (Addition on $\mathbb{R}$ Is Associative)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Sum of Cuts Is a Cut
- Addition on $\mathbb{Q}$ Is Associative

Theorem

Theorem 3.4.4 (Addition on $\mathbb{R}$ Is Commutative)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha + \beta = \beta + \alpha.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Sum of Cuts Is a Cut
- Addition on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 3.4.5 (Zero Is an Additive Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + 0_{\mathbb{R}} = \alpha.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Zero Is an Additive Identity on $\mathbb{Q}$

Definition

Definition 3.4.6 (Negation of a Cut). For $\alpha \in \mathbb{R}$, define

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q}\,(r > 0 \wedge -p - r \notin \alpha)\}.$$

Remark (Dependencies).

- The Real Numbers
- Rational Numbers
- Positive Rational
- Order on $\mathbb{Q}$

Lemma

Lemma 3.4.7 (Negation of a Cut Is a Cut). *If $\alpha \in \mathbb{R}$, then $-\alpha$ is a Dedekind cut.*

Remark (Dependencies).

- Negation on $\mathbb{R}$
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 3.4.8 (Negation Is an Additive Inverse on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + (-\alpha) = 0_{\mathbb{R}}.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Negation of a Cut Is a Cut

- Zero in $\mathbb{R}$

Theorem

Theorem 3.4.9 ($\mathbb{R}$ Is an Additive Abelian Group)**.** *The structure* $(\mathbb{R}, +, 0_{\mathbb{R}})$ *is an abelian group.*

Remark (Dependencies).

- Two-Sided Inverse
- Addition on $\mathbb{R}$
- Addition on $\mathbb{R}$ Is Associative
- Addition on $\mathbb{R}$ Is Commutative
- Zero Is an Additive Identity on $\mathbb{R}$
- Negation Is an Additive Inverse on $\mathbb{R}$

Definition

Definition 3.4.10 (Subtraction on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha - \beta := \alpha + (-\beta).$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Negation on $\mathbb{R}$

3.5 Multiplication on $\mathbb{R}$

Definition

Definition 3.5.1 (Nonnegative Cut)**.** A cut $\alpha \in \mathbb{R}$ is nonnegative if $0_{\mathbb{R}} \leq \alpha$ and positive if $0_{\mathbb{R}} < \alpha$.

Remark (Dependencies).

- The Real Numbers
- Zero in $\mathbb{R}$
- Order on $\mathbb{R}$

Definition

Definition 3.5.2 (Product of Nonnegative Cuts)**.** For nonnegative cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \cdot \beta := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\} \cup \{p \cdot q : p \in \alpha,\ q \in \beta,\ p \geq 0_{\mathbb{Q}},\ q \geq 0_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Nonnegative Cut
- Multiplication on $\mathbb{Q}$
- Order on $\mathbb{Q}$

Lemma

Lemma 3.5.3 (Product of Nonnegative Cuts Is a Cut). *If $\alpha, \beta \geq 0_{\mathbb{R}}$, then $\alpha \cdot \beta$ is a Dedekind cut.*

Remark (Dependencies).

- Product of Nonnegative Cuts
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Definition

Definition 3.5.4 (Absolute Value on $\mathbb{R}$). For $\alpha \in \mathbb{R}$, define

$$|\alpha| := \begin{cases} \alpha, & 0_{\mathbb{R}} \leq \alpha, \\ -\alpha, & \alpha < 0_{\mathbb{R}}. \end{cases}$$

Remark (Dependencies).

- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Negation on $\mathbb{R}$

Theorem

Theorem 3.5.5 (Triangle Inequality). *For all $\alpha, \beta \in \mathbb{R}$,*

$$|\alpha + \beta| \leq |\alpha| + |\beta|.$$

Go to proof.

Remark (Dependencies).

- Absolute Value on $\mathbb{R}$
- $\mathbb{R}$ Is an Ordered Field

Definition

Definition 3.5.6 (Multiplication of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define $\alpha \cdot \beta$ by sign cases: use $|\alpha| \cdot |\beta|$ when α and β have the same sign, use $-(|\alpha| \cdot |\beta|)$ when they have opposite signs, and use $0_{\mathbb{R}}$ when either factor is $0_{\mathbb{R}}$.

Remark (Dependencies).

- The Real Numbers
- Product of Nonnegative Cuts
- Absolute Value on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Zero in $\mathbb{R}$

Lemma

Lemma 3.5.7 (Product of Cuts Is a Cut)**.** *If* $\alpha, \beta \in \mathbb{R}$, *then* $\alpha \cdot \beta \in \mathbb{R}$.

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Nonnegative Cuts
- Product of Nonnegative Cuts Is a Cut
- Negation of a Cut Is a Cut

Theorem

Theorem 3.5.8 (Multiplication on $\mathbb{R}$ Is Associative)**.** *For all* $\alpha, \beta, \gamma \in \mathbb{R}$,

$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma).$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Cuts Is a Cut
- Multiplication on $\mathbb{Q}$ Is Associative

Theorem

Theorem 3.5.9 (Multiplication on $\mathbb{R}$ Is Commutative)**.** *For all* $\alpha, \beta \in \mathbb{R}$,

$$\alpha \cdot \beta = \beta \cdot \alpha.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Product of Cuts Is a Cut
- Multiplication on $\mathbb{Q}$ Is Commutative

Theorem

Theorem 3.5.10 (One Is a Multiplicative Identity on $\mathbb{R}$)**.** *For every* $\alpha \in \mathbb{R}$,

$$\alpha \cdot 1_{\mathbb{R}} = \alpha.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- One in $\mathbb{R}$
- One Is a Multiplicative Identity on $\mathbb{Q}$

Definition

Definition 3.5.11 (Reciprocal of a Positive Cut). For $\alpha > 0_{\mathbb{R}}$, define

$$\alpha^{-1} := \{r \in \mathbb{Q} : r \leq 0_{\mathbb{Q}}\} \cup \{r \in \mathbb{Q} : r > 0_{\mathbb{Q}} \wedge \exists s \in \mathbb{Q}\,(s > r \wedge s^{-1} \notin \alpha)\}.$$

Remark (Dependencies).

- The Real Numbers
- Order on $\mathbb{R}$
- Order on $\mathbb{Q}$
- Reciprocal on $\mathbb{Q}$

Definition

Definition 3.5.12 (Reciprocal of a Nonzero Cut). For $\alpha \neq 0_{\mathbb{R}}$, define α^{-1} to be $|\alpha|^{-1}$ if $\alpha > 0_{\mathbb{R}}$ and $-(|\alpha|^{-1})$ if $\alpha < 0_{\mathbb{R}}$.

Remark (Dependencies).

- Reciprocal of a Positive Cut
- Absolute Value on $\mathbb{R}$
- Negation on $\mathbb{R}$
- Zero in $\mathbb{R}$
- Order on $\mathbb{R}$

Lemma

Lemma 3.5.13 (Reciprocal of a Nonzero Cut Is a Cut). *If $\alpha \neq 0_{\mathbb{R}}$, then $\alpha^{-1} \in \mathbb{R}$.*

Remark (Dependencies).

- Reciprocal on $\mathbb{R}$
- Reciprocal of a Positive Cut
- Dedekind Cut
- $\mathbb{Q}$ Is an Ordered Field

Theorem

Theorem 3.5.14 (Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$). *If $\alpha \neq 0_{\mathbb{R}}$, then*

$$\alpha \cdot \alpha^{-1} = 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Reciprocal on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Reciprocal of a Nonzero Cut Is a Cut
- One in $\mathbb{R}$

Definition

Definition 3.5.15 (Division on $\mathbb{R}$). For $\beta \neq 0_{\mathbb{R}}$, define

$$\alpha/\beta := \alpha \cdot \beta^{-1}.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Reciprocal on $\mathbb{R}$
- Zero in $\mathbb{R}$

Theorem

Theorem 3.5.16 (Distributivity on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Multiplication Distributes over Addition on $\mathbb{Q}$

3.6 Field and Ordered-Field Structure of $\mathbb{R}$

Theorem

Theorem 3.6.1 (Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{R} \setminus \{0_{\mathbb{R}}\}, \cdot, 1_{\mathbb{R}})$ is an abelian group.*

Remark (Dependencies).

- Two-Sided Inverse

- Multiplication on $\mathbb{R}$
- Multiplication on $\mathbb{R}$ Is Associative
- Multiplication on $\mathbb{R}$ Is Commutative
- One Is a Multiplicative Identity on $\mathbb{R}$
- Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$

Theorem

Theorem 3.6.2 ($\mathbb{R}$ Is a Field)**.** *The structure* $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}})$ *is a field.*

Remark (Dependencies).

- $\mathbb{R}$ Is an Additive Abelian Group
- Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group
- Distributivity on $\mathbb{R}$
- Zero Is Not One in $\mathbb{R}$

Corollary

Corollary 3.6.3 (Field Laws Hold on $\mathbb{R}$)**.** *Every theorem proved generically for fields holds in* $\mathbb{R}$*, including zero product, double negation, sign rules, cancellation, uniqueness of inverses, inverse laws, and exponent laws.*

Remark (Dependencies).

- Two-Sided Inverse

Order Compatibility

Theorem

Theorem 3.6.4 (Addition Preserves Order on $\mathbb{R}$)**.** *For all* $\alpha, \beta, \gamma \in \mathbb{R}$*,*

$$\alpha < \beta \implies \alpha + \gamma < \beta + \gamma,$$

and the corresponding non-strict statement also holds.

Remark (Dependencies).

- Addition on $\mathbb{R}$
- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Addition Preserves Order on $\mathbb{Q}$

Theorem

Theorem 3.6.5 (Multiplication by Positives Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \gamma \wedge \alpha < \beta \implies \gamma \cdot \alpha < \gamma \cdot \beta.$$

Remark (Dependencies).

- Multiplication on $\mathbb{R}$
- Order on $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Positive Multiplication Preserves Order on $\mathbb{Q}$

Theorem

Theorem 3.6.6 ($\mathbb{R}$ Is an Ordered Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq)$ is an ordered field.*

Remark (Dependencies).

- $\mathbb{R}$ Is a Field
- $\mathbb{R}$ Is a Totally Ordered Set
- Addition Preserves Order on $\mathbb{R}$
- Multiplication by Positives Preserves Order on $\mathbb{R}$
- Trichotomy

Corollary

Corollary 3.6.7 (Ordered-Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for ordered fields holds in $\mathbb{R}$.*

Remark (Dependencies).

- Strict Total Order

Theorem

Theorem 3.6.8 ($\mathbb{R}$ Is a Complete Ordered Field). *The ordered field $\mathbb{R}$ satisfies the least-upper-bound property.*

Remark (Dependencies).

- Complete Ordered Field
- $\mathbb{R}$ Is an Ordered Field
- Least-Upper-Bound Property

3.7 Embedding $\mathbb{Q}$ into $\mathbb{R}$

Definition

Definition 3.7.1 (Embedding of $\mathbb{Q}$ into $\mathbb{R}$)**.** Define $\iota : \mathbb{Q} \to \mathbb{R}$ by

$$\iota(q) := q^*.$$

Remark (Dependencies).

- Rational Numbers
- The Real Numbers
- Rational Cut

Theorem

Theorem 3.7.2 (Embedding Preserves Zero and One)**.** *The embedding satisfies*

$$\iota(0_{\mathbb{Q}}) = 0_{\mathbb{R}} \qquad \textit{and} \qquad \iota(1_{\mathbb{Q}}) = 1_{\mathbb{R}}.$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Zero in $\mathbb{Q}$
- One in $\mathbb{Q}$
- Zero in $\mathbb{R}$
- One in $\mathbb{R}$

Theorem

Theorem 3.7.3 (Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective)**.** *For all* $p, q \in \mathbb{Q}$,

$$\iota(p) = \iota(q) \implies p = q.$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Rational Cut
- $\mathbb{Q}$ Is a Totally Ordered Set

Theorem

Theorem 3.7.4 (Embedding Preserves Addition and Multiplication)**.** *For all* $p, q \in \mathbb{Q}$,

$$\iota(p+q) = \iota(p) + \iota(q) \qquad \textit{and} \qquad \iota(p \cdot q) = \iota(p) \cdot \iota(q).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$

Theorem

Theorem 3.7.5 (Embedding Preserves Order)**.** *For all $p, q \in \mathbb{Q}$,*

$$p \leq q \iff \iota(p) \leq \iota(q).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Order on $\mathbb{Q}$
- Order on $\mathbb{R}$
- Rational Cut

Theorem

Theorem 3.7.6 ($\mathbb{Q}$ Embeds into $\mathbb{R}$ as an Ordered Field)**.** *The map ι is an injective order-preserving field homomorphism.*

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Embedding Preserves Zero and One
- Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective
- Embedding Preserves Addition and Multiplication
- Embedding Preserves Order

Theorem

Theorem 3.7.7 (Density of $\mathbb{Q}$ in $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha < \beta \implies \exists q \in \mathbb{Q}\,(\alpha < \iota(q) < \beta).$$

Remark (Dependencies).

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Strict Order on $\mathbb{R}$
- Density of $\mathbb{Q}$ in Itself

- Dedekind Cut

Theorem

Theorem 3.7.8 (Archimedean Property of $\mathbb{R}$). *For $\alpha, \beta \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \alpha \implies \exists n \in \mathbb{N}\,(\beta < \iota(n) \cdot \alpha).$$

Remark (Dependencies).

- Least-Upper-Bound Property

3.8 The Gap Filled

Definition

Definition 3.8.1 (The Square-Root-of-Two Cut). Define

$$\sqrt{2} := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}} \vee r \cdot r < 2_{\mathbb{Q}}\}.$$

Remark (Dependencies).

- Rational Numbers
- Order on $\mathbb{Q}$
- Multiplication on $\mathbb{Q}$
- Two in $\mathbb{Q}$

Lemma

Lemma 3.8.2 (The Square-Root-of-Two Cut Is a Cut). *The set $\sqrt{2}$ is a Dedekind cut.*

Remark (Dependencies).

- Density of $\mathbb{Q}$

Theorem

Theorem 3.8.3 (Its Square Is Two). *In $\mathbb{R}$,*

$$\sqrt{2} \cdot \sqrt{2} = 2_{\mathbb{R}},$$

where $2_{\mathbb{R}} := \iota(2_{\mathbb{Q}})$.

Remark (Dependencies).

- The Square-Root-of-Two Cut
- The Square-Root-of-Two Cut Is a Cut
- Multiplication on $\mathbb{R}$

- Embedding of $\mathbb{Q}$ into $\mathbb{R}$
- Two in $\mathbb{Q}$

Corollary

Corollary 3.8.4 ($\mathbb{R}$ Contains the Supremum $\mathbb{Q}$ Was Missing). *Let*

$$S = \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \wedge q \cdot q < 2_{\mathbb{Q}}\}.$$

Then

$$\sqrt{2} = \sup \iota(S).$$

Thus the bounded-above rational set with no rational supremum has a supremum in $\mathbb{R}$.

Remark (Dependencies).

- The Gap Set in $\mathbb{Q}$
- The Gap Set Has No Rational Supremum

3.9 Characterization of $\mathbb{R}$

Definition

Definition 3.9.1 (Complete Ordered Field). A *complete ordered field* is an ordered field satisfying the least-upper-bound property.

Remark (Dependencies).

- Strict Total Order
- Least-Upper-Bound Property on $\mathbb{R}$

Theorem

Theorem 3.9.2 (Uniqueness of the Complete Ordered Field). *Any two complete ordered fields are isomorphic by a unique order-isomorphism.*

Remark (Dependencies).

- Complete Ordered Field
- $\mathbb{R}$ Is a Complete Ordered Field

Corollary

Corollary 3.9.3 ($\mathbb{R}$ Is the Real Numbers). *The field* $\mathbb{R}$ *is determined up to unique order-isomorphism by being a complete ordered field.*

Remark (Dependencies).

- Complete Ordered Field
- Uniqueness of the Complete Ordered Field
- $\mathbb{R}$ Is a Complete Ordered Field

3.10 Transition and Forward

Remark 3.10.1 (What Completeness Buys). The least-upper-bound property promotes the arbitrary closeness available in $\mathbb{Q}$ to guaranteed limits in $\mathbb{R}$: bounded monotone sequences converge, Cauchy sequences converge, and suprema exist. The general theory of these phenomena belongs to analysis; this chapter supplies the complete ordered field in which they live.

Remark 3.10.2 (Parallel Constructions). Cauchy sequences, nested intervals, and dyadic expansions give alternative constructions of complete ordered fields. By Uniqueness of the Complete Ordered Field, each yields this same $\mathbb{R}$ up to unique isomorphism.

Chapter 4

Complex Numbers

complex-numbers

Real Numbers $\to$ **Complex Numbers**

4.1 Construction of $\mathbb{C}$

Complex Prepairs

The Last Number System Extension

The complex numbers extend the real numbers by adding a formal square root of -1. Unlike the construction of $\mathbb{Z}$ from $\mathbb{W}$ or $\mathbb{Q}$ from $\mathbb{Z}$, no nontrivial identification is needed: ordered pairs of real numbers already have the correct equality. We still use the same construction pattern: representatives, an equivalence relation, equivalence classes, then operations whose well-definedness is checked before they are used.

Definition

Definition 4.1.1 (Complex Prepair)**.** A *complex prepair* is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, read informally as the expression $a + bi$.

Remark (Dependencies).

- Ordered Pair
- The Real Numbers

Definition

Definition 4.1.2 (Complex Prepair Set)**.** The complex prepair set is

$$\operatorname{Pre}\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

Remark (Dependencies).

- Cartesian Product
- The Real Numbers

Coordinate Equivalence

Definition

Definition 4.1.3 (Complex Coordinate Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{C}$, define

$$(a, b) \sim_{\mathbb{C}} (c, d) \quad :\Longleftrightarrow \quad a = c \ \text{ and } \ b = d.$$

Remark (Predicate reading).

$$(a, b) \sim_{\mathbb{C}} (c, d) \Longleftrightarrow (a = c) \wedge (b = d).$$

Thus the complex-number quotient is formally present, but the equivalence classes are singleton coordinate-equality classes.

Remark (Dependencies).

- Relation
- Complex Prepair Set

Lemma

Lemma 4.1.4 (Complex Coordinate Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{C}$*,*

$$(a, b) \sim_{\mathbb{C}} (a, b).$$

Lemma

Lemma 4.1.5 (Complex Coordinate Equivalence Is Symmetric)**.** *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{C}} (a, b).$$

Lemma

Lemma 4.1.6 (Complex Coordinate Equivalence Is Transitive)**.** *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \wedge (c, d) \sim_{\mathbb{C}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{C}} (e, f).$$

Theorem

Theorem 4.1.7 (Complex Coordinate Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{C}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{C}$*.*

Remark (Proof sketch). Reflexivity, symmetry, and transitivity are checked coordinate by coordinate using equality in $\mathbb{R}$. Thus this quotient construction has the same formal shape as the constructions of $\mathbb{Z}$ and $\mathbb{Q}$, even though the equivalence classes are singleton classes.

Remark (Dependencies).

- Equivalence Relation
- Complex Coordinate Equivalence

- Complex Coordinate Equivalence Is Reflexive
- Complex Coordinate Equivalence Is Symmetric
- Complex Coordinate Equivalence Is Transitive

Definition of $\mathbb{C}$

Definition

Definition 4.1.8 (Complex Numbers)**.** The complex numbers are the quotient

$$\mathbb{C} := \operatorname{Pre}\mathbb{C}/{\sim_{\mathbb{C}}}.$$

Remark (Dependencies).

- Quotient Set
- Complex Prepair Set
- Complex Coordinate Equivalence

Definition

Definition 4.1.9 (Complex Class Notation)**.** The notation $[a, b]_{\mathbb{C}}$ denotes the $\sim_{\mathbb{C}}$-equivalence class of the prepair (a, b).

Remark (Standard form). Since coordinate equivalence is equality of coordinates, every complex number has a unique representative

$$[a, b]_{\mathbb{C}}.$$

We later write this same element as $a + bi$.

Definition

Definition 4.1.10 (Zero, One, and the Imaginary Unit in $\mathbb{C}$)**.** Define

$$0_{\mathbb{C}} := [0, 0]_{\mathbb{C}}, \qquad 1_{\mathbb{C}} := [1, 0]_{\mathbb{C}}, \qquad i := [0, 1]_{\mathbb{C}}.$$

Remark (Dependencies).

- Equivalence Class
- Complex Class Notation
- The Real Numbers

4.2 Operations on $\mathbb{C}$-Classes

Addition on $\mathbb{C}$-Classes

Definition

Definition 4.2.1 (Addition on $\mathbb{C}$-Classes)**.** Define addition on representatives by

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} := [a + c, b + d]_{\mathbb{C}}.$$

Remark (Dependencies).

- Complex Class Notation
- Addition on $\mathbb{R}$

Lemma

Lemma 4.2.2 (Addition Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ *and* $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$*, then*

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} + [c', d']_{\mathbb{C}}.$$

Remark (Dependencies).

- Addition on $\mathbb{C}$
- Complex Coordinate Equivalence

Theorem

Theorem 4.2.3 (Addition on $\mathbb{C}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Remark (Proof sketch). Suppose the input classes are represented in two ways. Coordinate equivalence gives $a = a'$, $b = b'$, $c = c'$, and $d = d'$. Addition in $\mathbb{R}$ gives $a + c = a' + c'$ and $b + d = b' + d'$, so the resulting complex classes agree.

Remark (Dependencies).

- Function
- Addition on $\mathbb{C}$
- Addition Respects Complex Coordinate Equivalence

Negation and Subtraction

Definition

Definition 4.2.4 (Negation on $\mathbb{C}$-Classes)**.** Define negation by

$$-[a, b]_{\mathbb{C}} := [-a, -b]_{\mathbb{C}}.$$

Lemma

Lemma 4.2.5 (Negation Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$*, then*

$$-[a, b]_{\mathbb{C}} = -[a', b']_{\mathbb{C}}.$$

Theorem

Theorem 4.2.6 (Negation on $\mathbb{C}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{C} \to \mathbb{C}$*.*

Remark (Dependencies)*.*

- Function
- Negation on $\mathbb{C}$
- Negation Respects Complex Coordinate Equivalence

Remark (Proof sketch)*.* If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$, then $a = a'$ and $b = b'$. Hence $-a = -a'$ and $-b = -b'$, so the negated representatives determine the same complex class.

Definition

Definition 4.2.7 (Subtraction on $\mathbb{C}$)**.** For $z, w \in \mathbb{C}$, define

$$z - w := z + (-w).$$

Theorem

Theorem 4.2.8 (Subtraction on $\mathbb{C}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$*.*

Remark (Dependencies)*.*

- Function
- Subtraction on $\mathbb{C}$
- Addition on $\mathbb{C}$ Is Well-Defined
- Negation on $\mathbb{C}$ Is Well-Defined

Multiplication on $\mathbb{C}$-Classes

Definition

Definition 4.2.9 (Multiplication on $\mathbb{C}$-Classes)**.** Define multiplication on representatives by

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} := [ac - bd, ad + bc]_{\mathbb{C}}.$$

Remark (Why this formula)*.* The rule is forced by the intended identity $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Remark (Dependencies).

- Complex Class Notation
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Lemma

Lemma 4.2.10 (Multiplication Respects Complex Coordinate Equivalence)**.** *If* $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ *and* $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$*, then*

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} \cdot [c', d']_{\mathbb{C}}.$$

Theorem

Theorem 4.2.11 (Multiplication on $\mathbb{C}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Remark (Proof sketch). Changing representatives preserves each coordinate. Therefore the real expressions $ac - bd$ and $ad + bc$ are unchanged after replacing (a, b) by an equivalent prepair and (c, d) by an equivalent prepair. The product class is independent of the chosen representatives.

Remark (Dependencies).

- Function
- Multiplication on $\mathbb{C}$
- Multiplication Respects Complex Coordinate Equivalence

Conjugation and Inverses

Definition

Definition 4.2.12 (Complex Conjugation)**.** For $z = [a, b]_{\mathbb{C}}$, define its *complex conjugate* by

$$\overline{z} := [a, -b]_{\mathbb{C}}.$$

Theorem

Theorem 4.2.13 (Complex Conjugation Is Well-Defined)**.** *Complex conjugation determines a well-defined unary operation* $\mathbb{C} \to \mathbb{C}$*.*

Remark (Dependencies).

- Function
- Complex Conjugation

Remark (Proof sketch). Equivalent prepairs have equal first coordinates and equal second coordinates. Keeping the first coordinate and negating the second preserves that equality, so conjugation does not depend on the representative.

Lemma

Lemma 4.2.14 (Product with the Conjugate Is Real and Nonnegative). *For* $z = [a, b]_{\mathbb{C}}$,

$$z\overline{z} = [a^2 + b^2, 0]_{\mathbb{C}}.$$

Moreover, if $z \neq 0_{\mathbb{C}}$, *then* $a^2 + b^2 \neq 0$.

Definition

Definition 4.2.15 (Multiplicative Inverse in $\mathbb{C}$). For $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, define

$$z^{-1} := \left[\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}\right]_{\mathbb{C}}.$$

Theorem

Theorem 4.2.16 (Inversion on $\mathbb{C}^\times$ Is Well-Defined). *Inversion determines a well-defined operation on the nonzero complex numbers.*

Remark (Dependencies).

- Function
- Multiplicative Inverse in $\mathbb{C}$
- Product with the Conjugate Is Real and Nonnegative

Remark (Proof sketch). If $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, then a and b are not both zero, so $a^2 + b^2 \neq 0$ in $\mathbb{R}$. Equivalent representatives have the same coordinates, hence the same denominator and the same inverse coordinates.

4.3 Field Structure of $\mathbb{C}$

Additive Structure

Theorem

Theorem 4.3.1 (Addition on $\mathbb{C}$ Is Associative). *For all* $z, w, u \in \mathbb{C}$,

$$(z + w) + u = z + (w + u).$$

Theorem

Theorem 4.3.2 (Addition on $\mathbb{C}$ Is Commutative). *For all* $z, w \in \mathbb{C}$,

$$z + w = w + z.$$

Theorem

Theorem 4.3.3 (Zero Is an Additive Identity on $\mathbb{C}$)**.** *For every $z \in \mathbb{C}$,*

$$0_{\mathbb{C}} + z = z \qquad \text{and} \qquad z + 0_{\mathbb{C}} = z.$$

Theorem

Theorem 4.3.4 (Every Complex Number Has an Additive Inverse)**.** *For every $z \in \mathbb{C}$,*

$$z + (-z) = 0_{\mathbb{C}} \qquad \text{and} \qquad (-z) + z = 0_{\mathbb{C}}.$$

Multiplicative Structure

Theorem

Theorem 4.3.5 (Multiplication on $\mathbb{C}$ Is Associative)**.** *For all $z, w, u \in \mathbb{C}$,*

$$(zw)u = z(wu).$$

Theorem

Theorem 4.3.6 (Multiplication on $\mathbb{C}$ Is Commutative)**.** *For all $z, w \in \mathbb{C}$,*

$$zw = wz.$$

Theorem

Theorem 4.3.7 (One Is a Multiplicative Identity on $\mathbb{C}$)**.** *For every $z \in \mathbb{C}$,*

$$1_{\mathbb{C}} z = z \qquad \text{and} \qquad z 1_{\mathbb{C}} = z.$$

Theorem

Theorem 4.3.8 (Every Nonzero Complex Number Has a Multiplicative Inverse)**.** *For every $z \in \mathbb{C}$ with $z \neq 0_{\mathbb{C}}$,*

$$zz^{-1} = 1_{\mathbb{C}} \qquad \text{and} \qquad z^{-1}z = 1_{\mathbb{C}}.$$

Distributivity and the Field Theorem

Theorem

Theorem 4.3.9 (Multiplication Distributes over Addition on $\mathbb{C}$)**.** *For all $z, w, u \in \mathbb{C}$,*

$$z(w + u) = zw + zu \qquad \text{and} \qquad (w + u)z = wz + uz.$$

Theorem

Theorem 4.3.10 ($\mathbb{C}$ Is a Field)**.** *With the operations defined above,*

$$(\mathbb{C}, +, \cdot, 0_{\mathbb{C}}, 1_{\mathbb{C}})$$

is a field.

Remark (Dependencies).

- Addition on $\mathbb{C}$ Is Well-Defined
- Multiplication on $\mathbb{C}$ Is Well-Defined
- Every Nonzero Complex Number Has a Multiplicative Inverse

The Imaginary Unit

Theorem

Theorem 4.3.11 (The Imaginary Unit Squares to Negative One)**.** *In* $\mathbb{C}$*,*

$$i^2 = -1_{\mathbb{C}}.$$

Remark (Interpretation)*.* The equation $i^2 = -1$ is the structural reason for introducing $\mathbb{C}$. It makes the polynomial $x^2 + 1$ have a root, while the field operations remain compatible with the real arithmetic already constructed.

Theorem

Theorem 4.3.12 ($\mathbb{C}$ Is Not an Ordered Field)**.** *There is no order relation on* $\mathbb{C}$ *making* $\mathbb{C}$ *an ordered field and extending the usual order on* $\mathbb{R}$*.*

Remark (Reason)*.* In any ordered field, every square is nonnegative. If $\mathbb{C}$ were an ordered field extending the real order, i^2 would be nonnegative. But $i^2 = -1$, and $-1 < 0$ in the real subfield.

4.4 Embedding $\mathbb{R}$ into $\mathbb{C}$

The Real Axis

Definition

Definition 4.4.1 (Canonical Embedding of $\mathbb{R}$ into $\mathbb{C}$)**.** Define

$$\iota : \mathbb{R} \to \mathbb{C}, \qquad \iota(a) := [a, 0]_{\mathbb{C}}.$$

Remark (Dependencies).

- The Real Numbers
- Complex Numbers

- Complex Class Notation

Theorem

Theorem 4.4.2 (The Real Embedding into $\mathbb{C}$ Is Injective). *If $\iota(a) = \iota(b)$, then $a = b$.*

Theorem

Theorem 4.4.3 (The Real Embedding Preserves Addition and Multiplication). *For all $a, b \in \mathbb{R}$,*

$$\iota(a + b) = \iota(a) + \iota(b), \qquad \iota(ab) = \iota(a)\iota(b).$$

Theorem

Theorem 4.4.4 (The Real Embedding Preserves Zero and One).

$$\iota(0) = 0_{\mathbb{C}}, \qquad \iota(1) = 1_{\mathbb{C}}.$$

Remark (Identification). After these preservation results, it is standard to identify $a \in \mathbb{R}$ with $[a, 0]_{\mathbb{C}} \in \mathbb{C}$. Under this convention,

$$[a, b]_{\mathbb{C}} = [a, 0]_{\mathbb{C}} + [0, b]_{\mathbb{C}} = a + bi.$$

Real and Imaginary Parts

Definition

Definition 4.4.5 (Real and Imaginary Parts). For $z = [a, b]_{\mathbb{C}}$, define

$$\mathrm{Re}(z) := a, \qquad \mathrm{Im}(z) := b.$$

Theorem

Theorem 4.4.6 (Real and Imaginary Parts Are Well-Defined). *The functions*

$$\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$$

are well-defined.

Remark (The completed number-system ladder). The construction of $\mathbb{C}$ completes the standard ladder of number systems in this volume:

$$\mathbb{N} \subseteq \mathbb{W} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

The final step sacrifices order compatibility in exchange for algebraic power: -1 now has a square root.

Chapter 5

Number Lines and Intervals

number-lines

Embedding the Number Systems → **Number Lines and Intervals** → Structure of the Real Line

5.1 Number Lines

The Geometric Model of an Ordered System

A number line is the geometric realization of a linearly ordered system: points on a line arranged so that "left of" matches the order. The construction so far is algebraic; the number line is the picture that accompanies it, and it becomes exact for $\mathbb{R}$.

Definition (Number Line)

Definition 5.1.1 (Number Line)**.** A *number line* for a linearly ordered system $(A, <_A)$ is a linearly ordered set of points L together with an order isomorphism $\lambda : A \to L$, so that for all $a, a' \in A$,

$$a <_A a' \iff \lambda(a) \text{ lies to the left of } \lambda(a').$$

Remark (Standard quantified statement).

$$\lambda : A \to L \text{ is a bijection with } \forall a, a' \in A \; (a <_A a' \Leftrightarrow \lambda(a) \prec \lambda(a')).$$

Remark (Interpretation). The number line records only order, not arithmetic: it is the visual home of betweenness and length. For $\mathbb{R}$ the order isomorphism is onto a full continuum, which is why the real line has no gaps; for $\mathbb{Q}$ the same picture has holes at every irrational location.

Remark (Dependencies).

- Order Embedding

5.2 Intervals on a Number Line

The Basic Pieces of the Line

Intervals are the natural named pieces of a number line. This section gives each basic interval shape its notation and spells out which endpoints are included.

Definition (Open Interval)

Definition 5.2.1 (Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *open interval* with endpoints a and b is

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Remark (Standard quantified statement).

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Remark (Interpretation)*.* The open interval excludes both endpoints. It is one of the four bounded interval shapes.

Remark (Dependencies).

- Strict Total Order

Definition (Closed Interval)

Definition 5.2.2 (Closed Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *closed interval* with endpoints a and b is

$$[a, b] = \{\, x \in \mathbb{R} : a \leq x \leq b \,\}.$$

Remark (Standard quantified statement).

$$[a, b] = \{\, x \in \mathbb{R} : a \leq x \leq b \,\}.$$

Remark (Interpretation)*.* The closed interval includes both endpoints. It is one of the four bounded interval shapes.

Remark (Dependencies).

- Strict Total Order

Definition (Left-Half-Open Interval)

Definition 5.2.3 (Left-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *left-half-open interval* with endpoints a and b is

$$(a, b] = \{\, x \in \mathbb{R} : a < x \leq b \,\}.$$

Remark (Standard quantified statement).

$$(a, b] = \{\, x \in \mathbb{R} : a < x \leq b \,\}.$$

Remark (Interpretation)*.* The left-half-open interval excludes the left endpoint, includes the right. It is one of the four bounded interval shapes.

Remark (Dependencies).

- Strict Total Order

Definition (Right-Half-Open Interval)

Definition 5.2.4 (Right-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *right-half-open interval* with endpoints a and b is

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Remark (Standard quantified statement).

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Remark (Interpretation). The right-half-open interval includes the left endpoint, excludes the right. It is one of the four bounded interval shapes.

Remark (Dependencies).

- Strict Total Order

Definition (Ray)

Definition 5.2.5 (Ray)**.** Let $a \in \mathbb{R}$. The four *rays* determined by a are

$$(a, \infty) = \{x : x > a\},\ [a, \infty) = \{x : x \geq a\},\ (-\infty, a) = \{x : x < a\},\ (-\infty, a] = \{x : x \leq a\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} (a, \infty) &= \{\, x \in \mathbb{R} : a < x \,\}, & [a, \infty) &= \{\, x \in \mathbb{R} : a \leq x \,\}, \\ (-\infty, a) &= \{\, x \in \mathbb{R} : x < a \,\}, & (-\infty, a] &= \{\, x \in \mathbb{R} : x \leq a \,\}. \end{aligned}$$

Remark (Interpretation). Rays are the unbounded one-sided intervals. Together with the bounded intervals and the whole line $(-\infty, \infty) = \mathbb{R}$, they exhaust the interval shapes.

Remark (Dependencies).

- Strict Total Order

Chapter 6

Summary: The Number Systems

numbers-summary

reals $\to$ **numbers-summary** $\to$ Finite Modular Number Systems

6.1 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

Derived Properties — Quick Reference

Name	Statement	Proof sketch
Zero multiplication	$a \cdot 0 = 0$	$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$; cancel $a \cdot 0$ via A4.
Negation rule	$(-a) \cdot b = -(ab)$	$ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$.
(-1) rule	$(-1) \cdot a = -a$	Special case: $b = a$, $-a = (-1)a$.
Double negation	$(-a)(-b) = ab$	$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.
No zero divisors	$ab = 0 \Rightarrow a = 0$ or $b = 0$	$\mathbb{Z}$ is an integral domain; proved via well-ordering + sign analysis.

The Natural Numbers $\mathbb{N}$

One-Based Peano Axioms for $\mathbb{N}$

Let $\mathbb{N}$ be a set with distinguished element 1 and successor operation $n \mapsto n{+\!+}$.

P1 $1 \in \mathbb{N}$.

P2 $n \in \mathbb{N} \Rightarrow n{+\!+} \in \mathbb{N}$. *(successor closure)*

P3 $n{+\!+} \neq 1$ for all $n \in \mathbb{N}$. *(the first element has no predecessor)*

P4 $n{+\!+} = m{+\!+} \Rightarrow n = m$. *(successor is injective)*

P5 If $P(0)$ holds and $P(n) \Rightarrow P(n{+\!+})$, then $P(n)$ holds for all $n \in \mathbb{N}$. *(induction)*

Remark. P1–P4 rule out wrap-around, ceiling, and rogue elements. P5 eliminates anything unreachable from 1 by finite iteration of the successor. Addition, multiplication, and order are not axioms – they are *defined* recursively from the successor and then shown to satisfy familiar laws by induction.

Remark (Well-ordering and induction are equivalent over $\mathbb{N}$)*.* The following are logically equivalent:

- Principle of Mathematical Induction (P5).
- Well-Ordering Principle: every nonempty $A \subseteq \mathbb{N}$ has a least element.
- Strong Induction: $(\forall k < n,\ P(k)) \Rightarrow P(n)$ for all n implies P holds everywhere.

One must assume at least one; the others follow. Well-ordering is the engine behind the Division Algorithm, Bézout's Identity, and unique prime factorization.

The Whole Numbers $\mathbb{W}$

$\mathbb{W}$ as $\mathbb{N}$ with an Adjoined Zero

The whole numbers are

$$\mathbb{W} = \mathbb{N} \cup \{0\}, \qquad 0 \notin \mathbb{N}.$$

The successor, addition, multiplication, and order on $\mathbb{W}$ extend the corresponding structures on $\mathbb{N}$:

- $S_{\mathbb{W}}(0) = 1$, and $S_{\mathbb{W}}$ agrees with $S_{\mathbb{N}}$ on $\mathbb{N}$;
- 0 is the additive identity;
- 0 is absorbing for multiplication;
- 0 is the least element of $\mathbb{W}$.

Remark (What $\mathbb{W}$ gains over $\mathbb{N}$, and what it lacks)*.* **Gain:** an additive identity. This makes $(\mathbb{W}, +, 0)$ a commutative monoid and prepares the ordered-pair construction of the integers.

Lack: additive inverses for nonzero elements. Subtraction is still not an internal operation in general, so $\mathbb{W}$ is not a ring.

The Integers $\mathbb{Z}$

Axioms for $\mathbb{Z}$ — Ordered Commutative Ring

Addition

A1	$a + b = b + a$	*(commutativity)*
A2	$(a + b) + c = a + (b + c)$	*(associativity)*
A3	$\exists\, 0 \in \mathbb{Z}$ such that $a + 0 = a$	*(additive identity)*
A4	$\forall\, a,\ \exists\, -a \in \mathbb{Z}$ with $a + (-a) = 0$	*(additive inverses)*

Multiplication

M1	$a \cdot b = b \cdot a$	*(commutativity)*
M2	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$	*(associativity)*
M3	$\exists\, 1 \in \mathbb{Z},\ 1 \neq 0$, such that $a \cdot 1 = a$	*(multiplicative identity)*

Distributivity

D	$a \cdot (b + c) = a \cdot b + a \cdot c$	*(distributivity)*

Order

O1	$a \leq b$ or $b \leq a$	*(totality)*
O2	$a \leq b \wedge b \leq a \Rightarrow a = b$	*(antisymmetry)*
O3	$a \leq b \wedge b \leq c \Rightarrow a \leq c$	*(transitivity)*
O4	$b \leq c \Rightarrow a + b \leq a + c$	*(order + addition)*
O5	$a \geq 0 \wedge b \geq 0 \Rightarrow ab \geq 0$	*(order + multiplication)*

Remark (What $\mathbb{Z}$ gains over $\mathbb{W}$, and what it lacks). **Gain:** Axiom A4 – additive inverses. Subtraction is always defined in $\mathbb{Z}$; it is not in $\mathbb{W}$.

Lack: Multiplicative inverses. There is no $x \in \mathbb{Z}$ with $2x = 1$, so M4 (every nonzero element is invertible) fails. In the language of abstract algebra, $\mathbb{Z}$ is a *commutative ordered ring* but not a field. The passage to $\mathbb{Q}$ restores multiplicative inverses by formally adjoining fractions.

Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Remark (Zero multiplication — the step most often looked up). The identity $a \cdot 0 = 0$ is not an axiom; it is derived. The proof uses only A3, A4, and D:

$$a \cdot 0 = a \cdot (0 + 0) \stackrel{\text{D}}{=} a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides (A4) gives $0 = a \cdot 0$, i.e. $a \cdot 0 = 0$. No order axioms, no well-ordering, no properties of $\mathbb{N}$ are needed — this holds in any ring.

Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Axiom Comparison Table

Group	**Axiom**	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$
Addition	A1 commutativity	✓	✓	✓	✓
	A2 associativity	✓	✓	✓	✓
	A3 identity (0)	×	✓	✓	✓
	A4 inverses ($-a$)	×	×	✓	✓
Multiplication	M1 commutativity	✓	✓	✓	✓
	M2 associativity	✓	✓	✓	✓
	M3 identity (1)	✓	✓	✓	✓
	M4 inverses ($1/a$)	×	×	×	✓
	D distributivity	✓	✓	✓	✓
Order	O1–O3 total order	✓	✓	✓	✓
	O4–O5 order + arithmetic	✓	✓	✓	✓
	Well-ordering	✓	✓	×	×
Completeness	Least upper bound	×	×	×	×
Structure		*positive arithmetic*	*comm. semiring*	*comm. ring*	*ordered*

✓ = satisfied × = fails

Remark (The extension hierarchy). Each number system repairs exactly one deficiency of its predecessor:

$$\mathbb{N} \xrightarrow{+\,0} \mathbb{W} \xrightarrow{+\,\text{A4}} \mathbb{Z} \xrightarrow{+\,\text{M4}} \mathbb{Q} \xrightarrow{+\,\text{LUB}} \mathbb{R}.$$

$\mathbb{N} \to \mathbb{W}$: adjoin 0 so addition has an identity. $\mathbb{W} \to \mathbb{Z}$: adjoin additive inverses so subtraction is always defined. $\mathbb{Z} \to \mathbb{Q}$: adjoin multiplicative inverses so division by any nonzero element is always defined. $\mathbb{Q} \to \mathbb{R}$: adjoin least upper bounds so every bounded increasing sequence converges.

Note that $\mathbb{N}$ and $\mathbb{W}$ are the well-ordered systems in the chain. $\mathbb{Z}$ loses well-ordering the moment negative integers are introduced (the set of all negative integers has no least element).

From Extension to Embedding

The hierarchy above describes what each new system adds. The next chapter looks at a separate question: how the earlier systems are identified inside the later ones. That identification is not merely a matter of writing subset symbols. It is carried by structure-preserving embeddings that justify treating a number from an earlier construction as the corresponding number in a larger system.

Bibliography